Volume-Independent Spectral Gap for Strong-Coupling Lattice Gauge Theory via Poincaré–Dobrushin Machinery

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Abstract

We establish the existence of a volume-independent mass gap $\Delta > 0$ for Wilson lattice gauge theory with compact simple gauge group G in the strong-coupling regime ($\beta < \beta_0$). For the continuum theory on $\mathbb{R}^4$, we record the conditional Osterwalder–Schrader reconstruction step: exponential clustering with uniform physical normalization would imply a continuum mass gap. Our approach constructs a renormalized free-energy functional $\mathcal{F}[\mu]$ over the lattice-regularized gauge field measure μ and proves its strict convexity in the regime where the Dobrushin–Zegarlinski estimate applies. The argument circumvents direct operator estimates on the Hamiltonian by reducing the lattice mass gap to a Poincaré inequality for the lattice gauge measure: the Holley–Stroock bounded perturbation lemma controls single-link conditional measures, and the Dobrushin–Zegarlinski criterion ensures that the resulting spectral gap is volume-independent. Via the transfer operator and reflection positivity, this Poincaré inequality translates into exponential clustering with a volume-independent rate.

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Contents

1 Introduction

The Yang–Mills mass gap problem, as formulated by Jaffe and Witten [1], asks whether for any compact simple gauge group G, the quantum Yang–Mills theory on $\mathbb{R}^4$ exists (in the sense of the Wightman axioms) and possesses a mass gap $\Delta > 0$: the lowest eigenvalue of the mass operator $M = \sqrt{P_\mu P^\mu}$ acting on the physical Hilbert space $\mathcal{H}$, restricted to the orthogonal complement of the vacuum, satisfies

$$\inf \sigma(M)\Big|_{\mathcal{H}\ominus\mathbb{C}\Omega} = \Delta > 0. \tag{1}$$

The classical Yang–Mills action on Euclidean $\mathbb{R}^4$ reads

$$S_{\mathrm{YM}}[A] = \frac{1}{2g^2}\int_{\mathbb{R}^4} \mathrm{tr}\left(F_{\mu\nu}F^{\mu\nu}\right) d^4x, \tag{2}$$

where $A = A_\mu^a T^a dx^\mu$ is the gauge connection taking values in the Lie algebra $\mathfrak{g}$, $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu]$ is the field-strength tensor, and g is the coupling constant.

Numerical evidence from lattice QCD computations strongly supports the existence of a mass gap [2, 10]. Constructive quantum field theory has established the existence of interacting theories in lower dimensions [6], and Balaban's renormalization group analysis on the lattice [7, 8, 9] provides detailed control over the ultraviolet structure. Chatterjee [23] has formulated key aspects of the Yang–Mills mass gap problem as problems in probability theory, providing a probabilistic framework closely related to the present approach. However, a complete proof combining ultraviolet regularity, the infinite-volume limit, and spectral positivity has remained elusive.

In this paper, we pursue a variational strategy. Rather than attempting to bound the Hamiltonian from below by direct operator inequalities, we construct a free-energy functional $\mathcal{F}[\mu]$ over probability measures on the space of lattice gauge configurations and demonstrate:

1. $\mathcal{F}$ is strictly convex, with a unique minimizer μ_* (the lattice vacuum state).

2. A Poincaré inequality for the lattice gauge measure μ_* holds with a volume-independent constant $\kappa > 0$: $\mathrm{Var}_{\mu_*}(f) \leq \kappa^{-1}\mathcal{E}(f, f)$ for all gauge-invariant f.

3. Via the transfer operator and reflection positivity, this Poincaré inequality yields exponential clustering and a mass gap $\Delta \geq \kappa > 0$ in the physical Hilbert space.

This paper is largely self-contained, relying on standard tools from functional analysis and probability theory (Holley–Stroock, Dobrushin–Zegarlinski, Martinelli, Guionnet–Zegarlinski) whose precise statements are recalled where needed. Connections to the broader FST programme are cited for context but are not logically required.

2 Lattice Regularization and the Free-Energy Functional

2.1 Wilson's Lattice Formulation

We regularize the theory on a finite hypercubic lattice $\Lambda = (a\mathbb{Z})^4 \cap [-L, L]^4$ with lattice spacing $a > 0$ and box size $L > 0$. The dynamical variables are group-valued link variables

$U_\ell \in G$ assigned to each oriented link ℓ of Λ. The Wilson action is

$$S_W[U] = \beta \sum_{p \in \mathcal{P}(\Lambda)} \left(1 - \frac{1}{N} \operatorname{Re} \operatorname{tr} U_p\right), \tag{3}$$

where $\beta = 2N/g^2$ (for $G = SU(N)$), $\mathcal{P}(\Lambda)$ is the set of elementary plaquettes, and $U_p = U_{\ell_1} U_{\ell_2} U_{\ell_3}^{-1} U_{\ell_4}^{-1}$ is the ordered product of link variables around the plaquette p.

The lattice partition function is

$$Z_\Lambda = \int_{G^{|\mathcal{L}(\Lambda)|}} e^{-S_W[U]} \prod_{\ell \in \mathcal{L}(\Lambda)} dU_\ell, \tag{4}$$

where dU_ℓ is the normalized Haar measure on G and $\mathcal{L}(\Lambda)$ denotes the set of links.

Definition 2.1 (Lattice gauge measure)**.** The lattice gauge measure is the probability measure

$$d\mu_\Lambda[U] = \frac{1}{Z_\Lambda} e^{-S_W[U]} \prod_\ell dU_\ell. \tag{5}$$

2.2 The Free-Energy Functional

We define the free-energy functional over probability measures μ on the configuration space $\mathcal{C}_\Lambda = G^{|\mathcal{L}(\Lambda)|}$.

Definition 2.2 (Free-energy functional)**.** For a probability measure μ on $\mathcal{C}_\Lambda$, absolutely continuous with respect to the Haar product measure $\lambda = \prod_\ell dU_\ell$, we define

$$\mathcal{F}[\mu] = \int_{\mathcal{C}_\Lambda} S_W[U]\, d\mu[U] + T \int_{\mathcal{C}_\Lambda} \log \frac{d\mu}{d\lambda}[U]\, d\mu[U], \tag{6}$$

where the first term is the expected action (internal energy) and the second term is the negative entropy, scaled by $T > 0$. The parameter T plays the role of an auxiliary variational temperature and is *independent* of the lattice coupling $\beta = 2N/g^2$; the physical lattice measure is recovered at $T = 1$.

The functional $\mathcal{F}[\mu]$ decomposes as

$$\mathcal{F}[\mu] = \langle S_W \rangle_\mu + T \cdot D_{\mathrm{KL}}(\mu \| \lambda) - T \log \operatorname{vol}(\mathcal{C}_\Lambda), \tag{7}$$

where $D_{\mathrm{KL}}(\mu\|\lambda)$ is the Kullback–Leibler divergence. Since dU_ℓ is the normalised Haar measure ($\operatorname{vol}(G) = 1$), the last term vanishes; we include it for conceptual clarity. At $T = 1$ (the physical value), the unique minimizer of $\mathcal{F}$ is precisely μ_Λ.

Lemma 2.3 (Strict convexity)**.** *The functional $\mathcal{F}[\mu]$ defined in (6) is strictly convex on the space $\mathcal{P}_{\mathrm{ac}}(\mathcal{C}_\Lambda)$ of probability measures absolutely continuous with respect to λ. Specifically, for any $\mu_0, \mu_1 \in \mathcal{P}_{\mathrm{ac}}(\mathcal{C}_\Lambda)$ with $\mu_0 \neq \mu_1$ and any $\theta \in (0,1)$:*

$$\mathcal{F}[\theta\mu_0 + (1-\theta)\mu_1] < \theta\mathcal{F}[\mu_0] + (1-\theta)\mathcal{F}[\mu_1]. \tag{8}$$

Proof. The energy term $\mu \mapsto \langle S_W \rangle_\mu$ is linear and hence convex. The entropy term $\mu \mapsto \int \log(d\mu/d\lambda)\, d\mu$ is strictly convex, as it equals $D_{\mathrm{KL}}(\mu\|\lambda)$ up to an additive constant. The strict convexity of the KL divergence is a standard result following from the strict convexity of $x \mapsto x \log x$ on $(0, \infty)$; see e.g. [13], Theorem 2.7.2. The sum of a linear functional and a strictly convex functional is strictly convex. □

2.3 The Hessian of $\mathcal{F}$ and the Spectral Bound

To connect the curvature of $\mathcal{F}$ at its minimum to the mass gap, we study the second variation of $\mathcal{F}$ around the equilibrium measure μ_Λ.

Definition 2.4 (Second variation)**.** Let μ_Λ be the unique minimizer of $\mathcal{F}$, and let $\delta\mu$ be a signed measure with $\int d(\delta\mu) = 0$ (preserving normalization) and $\delta\mu \ll \mu_\Lambda$. Writing $\delta\mu = f \cdot \mu_\Lambda$ with $\langle f \rangle_{\mu_\Lambda} = 0$, the second variation is

$$\delta^2 \mathcal{F}[\mu_\Lambda](f, f) = T \int_{\mathcal{C}_\Lambda} f^2 \, d\mu_\Lambda = T \cdot \mathrm{Var}_{\mu_\Lambda}(f). \tag{9}$$

Remark 2.5*.* The Hessian (9) shows that the curvature of $\mathcal{F}$ at the minimum is controlled by the variance of perturbations under μ_Λ. For gauge-invariant perturbations, this variance is related to the connected two-point correlator of the gauge field.

Remark 2.6 (Primary gap object: Poincaré/LSI constant, not Hessian variance)*.* The Poincaré inequality constant κ (or equivalently the log-Sobolev constant α) is the primary gap object, not the Hessian variance. The Hessian $\nabla^2 \mathcal{F}[\mu_\Lambda]$ controls local curvature, but the spectral gap Δ is determined by the Poincaré constant via $\Delta = -\log(1 - \kappa) \geq \kappa$. The log-Sobolev constant $\alpha \geq \kappa/2$ provides the stronger exponential decay in relative entropy. In particular, the volume-independent mass gap (Theorem 4.1) is established through the Poincaré–Dobrushin machinery (Steps 1–2), not through the Hessian of $\mathcal{F}$ alone: the Hessian identity $\delta^2 \mathcal{F} = T \cdot \mathrm{Var}_{\mu_\Lambda}$ motivates the approach, but the quantitative gap bound $\Delta_\Lambda \geq \kappa_*$ comes from the Holley–Stroock and Dobrushin–Zegarlinski constants.

Remark 2.7 (Poincaré inequality vs. log-Sobolev inequality)*.* Throughout this paper, two functional inequalities appear:

(i) The **Poincaré inequality (PI)** states $\mathrm{Var}_\mu(f) \leq \kappa^{-1} \mathcal{E}(f, f)$; equivalently, κ is the spectral gap of the generator. PI controls the *rate of variance decay* (polynomial in time) under the associated Markov semigroup.

(ii) The **log-Sobolev inequality (LSI)** states $\mathrm{Ent}_\mu(f^2) \leq (2/\alpha) \, \mathcal{E}(f, f)$; the constant α controls *exponential decay of relative entropy* (hypercontractivity).

The hierarchy is strict: LSI with constant α implies PI with constant $\kappa \geq \alpha$ (Rothaus [Rothaus(1985)]), but PI does *not* imply LSI in general (counterexamples exist on heavy-tailed measures). In this paper, the core results (Theorem 4.1, Steps 1–2) establish a **Poincaré inequality**, not a log-Sobolev inequality. The LSI terminology appears in the discussion of the Bauerschmidt–Bodineau–Dagallier programme (Remark 5.21) and in the Kingman–Lyapunov framework (Section 5.11), where the stronger LSI would yield better quantitative bounds but is not required for the mass gap conclusion.

3 Reflection Positivity and the Transfer Operator

To pass from the Euclidean lattice theory to the physical Hilbert space, we employ the Osterwalder–Schrader reconstruction [3, 4].

3.1 Reflection Positivity on the Lattice

Consider the lattice Λ with a time reflection ϑ about the hyperplane $\{x_0 = 0\}$, acting on link variables by $\vartheta(U_\ell) = U_{\vartheta(\ell)}^{-1}$.

Lemma 3.1 (Reflection positivity of the lattice gauge measure)**.** *Let G be a compact Lie group and assume the plaquette weight $w : G \to \mathbb{R}$ admits a Peter–Weyl expansion*

$$w(g) = \sum_{\pi \in \widehat{G}} a_\pi \, \chi_\pi(g), \qquad a_\pi \geq 0, \tag{10}$$

where $\widehat{G}$ denotes the set of irreducible unitary representations and χ_π their characters. Then the lattice gauge measure μ_Λ satisfies the Osterwalder–Schrader reflection positivity condition: for any bounded measurable function f depending only on link variables in $\Lambda_+ = \Lambda \cap \{x_0 > 0\}$,

$$\langle \overline{\vartheta(f)} \cdot f \rangle_{\mu_\Lambda} \geq 0. \tag{11}$$

Proof. Decompose the plaquette set as $\mathcal{P}(\Lambda) = \mathcal{P}_+ \dot\cup \mathcal{P}_- \dot\cup \mathcal{P}_0$, where $\mathcal{P}_\pm$ consists of plaquettes contained entirely in $\Lambda_\pm$ and $\mathcal{P}_0$ consists of plaquettes crossing the reflection plane $\{x_0 = 0\}$. Writing the link variables as $U = (U_+, U_0, U_-)$ according to the induced partition of links, the density factorises as

$$\prod_{p \in \mathcal{P}(\Lambda)} w(U_p) = W_+(U_+, U_0) \, W_-(U_-, U_0) \, W_0(U_+, U_0, U_-),$$

where $W_\pm = \prod_{p \in \mathcal{P}_\pm} w(U_p)$ and $W_0 = \prod_{p \in \mathcal{P}_0} w(U_p)$.

Step 1: Reduction to kernel positivity. For $f \in \mathcal{A}_+$ (functions of U_+ only), fix U_0 and observe that (Here U_0 denotes the spatial links in the reflection hyperplane $\{x_0 = 0\}$, which belong to neither Λ_+ nor Λ_-.)

$$\langle \overline{\vartheta(f)} \cdot f \rangle_{\mu_\Lambda} = \frac{1}{Z_\Lambda} \int dU_0 \, \Big\langle f(\cdot, U_0), \; K_{U_0} \, f(\cdot, U_0) \Big\rangle_{L^2(dU_+)},$$

where $K_{U_0}(U_+, U_-) := \underline{W_0}(U_+, U_0, U_-)$ is the crossing kernel (after the change of variables $U_- \mapsto \vartheta(U_+)$ in the $\overline{\vartheta(f)}$-factor; this substitution preserves the Haar measure since $dU \mapsto d(U^{-1})$ is an invariant measure on any compact group). It suffices to show that K_{U_0} is a positive-definite kernel for each fixed U_0.

Step 2: Single-plaquette positivity via Peter–Weyl. Each crossing plaquette $p \in \mathcal{P}_0$ contains exactly one link $U_{+,p}$ in Λ_+ and the reflected link $U_{-,p}$ in Λ_-, so that $U_p = A_p(U_0) \, U_{+,p} \, B_p(U_0) \, U_{-,p}^{-1}$ for fixed group elements A_p, B_p depending on U_0.

By assumption (10), w is a central positive-definite function. The Peter–Weyl orthogonality relations give, for any $\phi \in L^2(G)$:

$$\iint \overline{\phi(g)} \, w(gh^{-1}) \, \phi(h) \, dg \, dh = \sum_\pi a_\pi \sum_{i,j} \left| \int \phi(g) \, \overline{\pi_{ij}(g)} \, dg \right|^2 \geq 0.$$

By Haar invariance, $(U_{+,p}, U_{-,p}) \mapsto w(A_p \, U_{+,p} \, B_p \, U_{-,p}^{-1})$ is therefore a positive-definite kernel.

Step 3: Product of positive-definite kernels. The crossing kernel is the product $K_{U_0} = \prod_{p \in \mathcal{P}_0} w(A_p \, U_{+,p} \, B_p \, U_{-,p}^{-1})$. Since pointwise products of positive-definite kernels are positive-definite (Schur product theorem), K_{U_0} is positive-definite, completing the proof. $\square$

Remark 3.2 (Wilson plaquette weight satisfies (10)). For the standard Wilson weight $w(g) = \exp\left(\frac{\beta}{N} \operatorname{Re}\operatorname{tr}\rho(g)\right)$ in the fundamental representation ρ, the character expansion coefficients $a_\pi(\beta)$ are the modified Bessel-type integrals $a_\pi(\beta) = \int_G w(g)\,\overline{\chi_\pi(g)}\,dg$, which are nonnegative for all $\beta \geq 0$ by a result of Osterwalder and Seiler [5]. Alternatively, the heat-kernel action $w_t(g) = \sum_\pi d_\pi\, e^{-t\,c_2(\pi)}\,\chi_\pi(g)$ has manifestly nonnegative coefficients for any compact G.

3.2 The Transfer Operator and Its Spectrum

Reflection positivity allows the construction of a transfer operator $\hat{T}$ acting on the physical Hilbert space $\mathcal{H}_{\text{phys}}$.

Definition 3.3 (Transfer operator)**.** The transfer operator $\hat{T} : \mathcal{H}_{\text{phys}} \to \mathcal{H}_{\text{phys}}$ is defined via the kernel

$$\langle f|\hat{T}|g\rangle = \int f(U_+)\,K(U_+, U_-)\,g(U_-)\prod_\ell dU_\ell, \tag{12}$$

where $K(U_+, U_-)$ is the transfer kernel obtained by integrating out temporal links connecting two adjacent time slices. The Hamiltonian is defined by $H = -\log\hat{T}$ (in lattice units $a = 1$).

The mass gap in the lattice theory is

$$\Delta_\Lambda = -\lim_{t\to\infty}\frac{1}{t}\log\frac{\langle\mathcal{O}(t)\mathcal{O}(0)\rangle_{\mu_\Lambda}}{\langle\mathcal{O}\rangle^2_{\mu_\Lambda}} \tag{13}$$

for any gauge-invariant observable $\mathcal{O}$ with $\langle\mathcal{O}\rangle_{\mu_\Lambda} \neq 0$. Equivalently, Δ_Λ is the gap between the two largest eigenvalues of $\hat{T}$:

$$\Delta_\Lambda = \log\frac{\lambda_0(\hat{T})}{\lambda_1(\hat{T})}, \tag{14}$$

where $\lambda_0 > \lambda_1 \geq \lambda_2 \geq \cdots$ are the eigenvalues of $\hat{T}$ in the gauge-invariant sector.

3.3 Reflection positivity under weak coupling

Lemma 3.1 establishes reflection positivity (RP) for the lattice Yang–Mills measure in the strong-coupling regime. Extending RP to the weak-coupling regime $\beta > \beta_0$ is essential for the Osterwalder–Schrader reconstruction.

Proposition 3.4 (RP preservation under block-spin RG)**.** *Let μ_Λ be a lattice Yang–Mills measure satisfying RP with respect to a hyperplane Π. If the block-spin kernel $\mathcal{R}$ preserves the reflection symmetry (i.e., $\mathcal{R}\circ\Theta = \Theta\circ\mathcal{R}$ where Θ is the reflection), then the blocked measure $\mu_{\Lambda'} = \mathcal{R}_*\mu_\Lambda$ also satisfies RP.*

Proof sketch. RP states that $\langle\Theta f, f\rangle_\mu \geq 0$ for all functions f supported on one side of Π. Since $\mathcal{R}$ commutes with Θ, the blocked function $\mathcal{R}f$ is supported on the corresponding side of the blocked hyperplane. The RP of μ' follows from:

$$\langle\Theta(\mathcal{R}f), \mathcal{R}f\rangle_{\mu'} = \langle\mathcal{R}(\Theta f), \mathcal{R}f\rangle_{\mu'} = \langle f, \mathcal{R}^*\Theta\mathcal{R}f\rangle_\mu \geq 0,$$

where the last inequality holds because $\Theta \geq 0$ (RP of μ) and $\mathcal{R}$ is a positive linear operator, so $\mathcal{R}^*\Theta\mathcal{R} \geq 0$ as a quadratic form. (Note: $\mathcal{R}$ is *not* an isometry; the argument uses positivity of the composed operator, not norm preservation.) The key condition is that $\mathcal{R}$ respects the lattice symmetries, which holds for standard block-spin transformations (Balaban, 1984 [8]). □

Remark 3.5 (Complete RP programme for the continuum limit). The proposition reduces the RP problem to verifying two conditions: (1) RP holds at the initial (strong-coupling) scale, which is Lemma 3.1; (2) the block-spin kernel commutes with reflections, which is a symmetry property of the RG scheme. Together with the gap transport conjecture (Conjecture 5.14), this provides a complete programme for constructing a reflection-positive continuum measure with a mass gap.

4 Main Theorem: Existence of the Mass Gap

We now state and prove the central result connecting the strict convexity of the free-energy functional to the mass gap.

Theorem 4.1 (Spectral gap from free-energy convexity)**.** *Let G be a compact simple Lie group and $\beta < \beta_0(d, G) := 1/C(d, G)$ (strong-coupling regime). The transfer operator $\hat{T}$ of the Wilson lattice gauge theory on Λ has a spectral gap: in the gauge-invariant sector of $\mathcal{H}_{\mathrm{phys}}$,*

$$\Delta_\Lambda = \log \frac{\lambda_0}{\lambda_1} \geq \kappa(\beta, G) > 0, \tag{15}$$

where $\kappa(\beta, G)$ depends on the coupling β and the gauge group G but is independent of the lattice volume $|\Lambda|$.

Proof. The proof combines three ingredients.

Step 1 (Single-link Poincaré inequality via bounded perturbation). For any gauge-invariant function $f \in L^2(\mu_\Lambda)$ with $\langle f\rangle_{\mu_\Lambda} = 0$:

$$\mathrm{Var}_{\mu_\Lambda}(f) \leq \frac{1}{\kappa} \cdot \mathcal{E}(f, f), \tag{16}$$

where $\mathcal{E}(f, f) = \sum_{\ell\in\mathcal{L}(\Lambda)} \langle (f - E_\ell[f])^2\rangle_{\mu_\Lambda}$ is the Dirichlet form of the heat-bath (Glauber) dynamics, with E_ℓ denoting the conditional expectation at link ℓ.

The constant κ is controlled by the single-link conditional measures. For a fixed link ℓ, the conditional density $\mu_\Lambda(dU_\ell \mid U_{\mathcal{L}\setminus\{\ell\}}) \propto \exp(-W(U_\ell))\, dU_\ell$, where $W(U) = -(\beta/N) \sum_{p\ni\ell} \mathrm{Re}\,\mathrm{tr}(A_p U)$ and the $A_p \in G$ depend on the neighbouring links. Since $|\mathrm{Re}\,\mathrm{tr}(A_p U)| \leq N$, the potential has bounded oscillation:

$$\mathrm{osc}(W) = \sup W - \inf W \leq 2\, d_\ell\, \beta, \tag{17}$$

where $d_\ell = 2(d - 1) = 6$ is the number of plaquettes containing ℓ in $d = 4$ dimensions on the hypercubic lattice. For $G = SU(N)$ in the fundamental representation, $|\mathrm{Re}\,\mathrm{tr}(A_pU)| \leq N$, so the bound (17) holds with the stated constant. For a general compact simple group G, the Wilson action is defined using a faithful representation ρ with normalised trace $\mathrm{tr}_\rho(\cdot)/\dim(\rho)$; the oscillation then satisfies $\mathrm{osc}(W) \leq 2\, d_\ell\, \beta\, C_\rho$, where $C_\rho := \sup_{g\in G} |\mathrm{tr}_\rho(g)|/\dim(\rho) \le 1$ is the normalised character bound. By the Holley–Stroock bounded perturbation lemma [14], the single-link conditional measure inherits a Poincaré inequality from the Haar measure on G with constant

$$\kappa_{\mathrm{link}}(\beta, G) \ge e^{-2\, d_\ell\, \beta\, C_\rho}, \tag{18}$$

since $\kappa_{\mathrm{Haar}}(G) = 1$ for the heat-bath dynamics on any compact group (a single complete resample from the Haar measure eliminates all variance). This bound is uniform in the boundary configuration and in $|\Lambda|$.

Step 2 (Volume independence via Dobrushin–Zegarlinski criterion). To ensure that κ does not vanish as $|\Lambda| \to \infty$, we verify the Dobrushin smallness condition for the Wilson lattice measure. Define the influence coefficients $c_{\ell,\ell'} \ge 0$ as the total-variation sensitivity of the conditional measure at link ℓ to changes at link ℓ'. Since the Wilson action has finite range (each plaquette couples at most 4 links), $c_{\ell,\ell'} = 0$ unless ℓ' shares a plaquette with ℓ. Moreover, for neighbouring links ℓ, ℓ' sharing $n_{\ell,\ell'}$ plaquettes (at most 2 in $d = 4$), the total-variation sensitivity admits a tanh-bound:

$$c_{\ell,\ell'} \le n_{\ell,\ell'} \cdot \tanh\Big(\beta\, C(G)/2\Big), \tag{19}$$

where $C(G) \le 1$ is the normalised character bound. (The conditional log-density at link ℓ depends on ℓ' only through plaquettes containing both links. Changing the staple A_p at a single plaquette shifts the conditional log-density by at most $\beta\, C(G)$. The total-variation distance between two probability densities $\propto e^{\pm h/2}$ with $\|h\|_\infty \le \beta\, C(G)$ is at most $\tanh(\beta\, C(G)/2)$—this is a standard bound; see e.g. [21].) Since each link has at most $d_\ell = 2(d-1)$ neighbouring plaquettes, the Dobrushin constant satisfies

$$\eta(\beta) := \sup_\ell \sum_{\ell'} c_{\ell,\ell'} \le d_\ell \cdot \tanh\Big(\beta\, C(G)/2\Big). \tag{20}$$

The Dobrushin uniqueness condition $\eta(\beta) < 1$ holds for $\beta < \beta_0 := 2\,\mathrm{artanh}(1/d_\ell\, C(G))$, and the Zegarlinski criterion [15] yields a volume-independent Poincaré inequality:

$$\kappa(\Lambda) \ge (1 - \eta(\beta))\, \kappa_{\mathrm{link}}(\beta, G) =: \kappa_* > 0, \tag{21}$$

independent of $|\Lambda|$. This establishes the lattice mass gap in the strong-coupling regime.

Remark 4.2 (Extension beyond strong coupling). The extension to all $\beta > 0$ is not a consequence of the cluster expansion, which converges only for $\beta < \beta_0$. While for any fixed finite lattice Λ the partition function $Z_\Lambda(\beta)$ is analytic in β (compact integration domain), this does not exclude phase transitions in the thermodynamic limit $|\Lambda| \to \infty$. In the strong-coupling regime, the uniform gap is rigorous; the persistence of a positive gap for all $\beta > 0$ in 4D $SU(N)$ gauge theory is a physically well-motivated conjecture (supported by lattice Monte Carlo data) but remains an open mathematical problem.

Remark 4.3 (Effective influence matrix). The Dobrushin interdependence matrix $C = (c_{ij})$ with entries

$$c_{ij} = \sup_{\omega,\omega' : \omega_k = \omega'_k \forall k \ne j} \|\mu_i(\cdot|\omega) - \mu_i(\cdot|\omega')\|_{\mathrm{TV}}$$

quantifies the influence of site j on the conditional distribution at site i. For lattice Yang–Mills theory, $c_{ij} = 0$ unless i and j share a plaquette, giving a sparse matrix with spectral radius

$$\rho(C) \le d_l \cdot \sup_l c_{l,\mathrm{nbr}} \le 6 \cdot \tanh(\beta/2).$$

The Dobrushin uniqueness condition $\rho(C) < 1$ holds for $\beta < 2\,\mathrm{artanh}(1/6) \approx 0.336$; this is the threshold used in the main theorem above. (A naïve linearisation $\tanh(x) \le x$ would give the weaker threshold $\beta_0 \approx 1/36 \approx 0.028$.)

On the block level (after one RG step), the effective influence matrix C' has entries bounded by $\rho(C)^{2^d}$ (geometric decay of correlations across blocks), yielding $\rho(C') \le \rho(C)^{16}$ in 4D. This exponential improvement per RG step is the mechanism behind Gap Transport.

Step 3 (Transfer operator spectral gap). Decompose the link variables as $(U(t), V(t))_{t=-T}^{T}$, where $U(t)$ collects the spatial links at time t and $V(t)$ the temporal links between slices t and $t+1$. The Wilson action splits accordingly into spatial contributions $S_{\mathrm{sp}}(U(t))$ and inter-slice contributions $S_{t,t+1}(U(t), V(t), U(t+1))$. Define the one-step transfer kernel by integrating out the temporal links:

$$K(U, W) := \int_{G^{E_{\mathrm{temp}}}} \exp\Big(-S_{t,t+1}(U, V, W)\Big)\, dV. \tag{22}$$

By the time-reflection symmetry of the Wilson action and the invariance of Haar measure, $K(U, W) = K(W, U)$. By reflection positivity (Lemma 3.1), K defines a positive self-adjoint operator on L^2.

Let $\pi(U) \propto e^{-S_{\mathrm{sp}}(U)} \int K(U, W)\, e^{-S_{\mathrm{sp}}(W)}\, dW$ be the slice marginal under μ_Λ (with periodic boundary conditions in time). The normalised Markov operator

$$(Pf)(U) := \frac{\int K(U, W)\, e^{-S_{\mathrm{sp}}(W)}\, f(W)\, dW}{\int K(U, W)\, e^{-S_{\mathrm{sp}}(W)}\, dW} \tag{23}$$

satisfies $P\mathbf{1} = \mathbf{1}$ and is reversible with respect to π. To verify detailed balance, write $Z(U) := \int K(U, W)\, e^{-S_{\mathrm{sp}}(W)}\, dW$ for the normalising denominator. Then

$$\pi(U)\, P(U, W) \;=\; \frac{1}{Z_{\mathrm{tot}}}\, e^{-S_{\mathrm{sp}}(U)}\, K(U, W)\, e^{-S_{\mathrm{sp}}(W)},$$

which is manifestly symmetric in (U, W) since $K(U, W) = K(W, U)$ and S_{sp} acts on each slice independently. Hence $\pi(U)\, P(U, W) = \pi(W)\, P(W, U)$, confirming that P is self-adjoint on $L^2(\pi)$. In particular, π is a stationary measure for P: integrating the detailed balance identity over W yields $\int \pi(W)\, P(W, U)\, dW = \int \pi(U)\, P(U, W)\, dW = \pi(U)$, i.e., $\pi P = \pi$.

The full-lattice Poincaré inequality (Step 2) implies exponential decay of correlations in the time direction, which yields the transfer-operator spectral gap. The argument follows the standard route from mixing to correlation decay for Gibbs measures satisfying the Dobrushin uniqueness condition; see Martinelli [16], Section 4.3 (Theorem 4.1 and Corollary 4.2 therein). While Martinelli's presentation focuses on finite-state spin systems, the Dobrushin uniqueness theory and its consequences for correlation decay extend to compact continuous state spaces (such as G-valued link variables) without essential modification: the Dobrushin–Shlosman theory [21] is formulated in terms of total-variation distances between conditional measures, which are defined for arbitrary probability spaces and do not require discreteness. The key ingredients—finite range, bounded conditional oscillation, and the Dobrushin comparison theorem—depend only on the product structure and compactness of the single-site state space. Guionnet and Zegarliński [17] further develop the stronger logarithmic Sobolev inequality (which implies the Poincaré inequality) for lattice systems with compact continuous state spaces.

In the Dobrushin uniqueness regime ($\eta(\beta) < 1$), the lattice Gibbs measure μ_Λ satisfies exponential decay of correlations: for any two bounded local functions f_1, f_2 supported on regions $A_1, A_2 \subset \Lambda$ with $\mathrm{dist}(A_1, A_2) = d$,

$$\left|\mathrm{Cov}_{\mu_\Lambda}(f_1, f_2)\right| \le C_1 \, \|f_1\|_\infty \, \|f_2\|_\infty \, |A_1| \, |A_2| \, e^{-d/\xi}, \tag{24}$$

where $\xi = O(1/\kappa_*)$ is the correlation length and C_1 depends only on the lattice dimension and the interaction range. This is a standard consequence of the Dobrushin comparison theorem [21, 16, 17]: the Dobrushin condition $\eta < 1$ controls the spatial decay of conditional expectations, which in turn controls correlations.

We apply (24) to gauge-invariant *local* slice observables $\mathcal{O}_1$ at time 0 and $\mathcal{O}_2$ at time t, where each $\mathcal{O}_i$ is supported on a fixed (volume-independent) set of spatial links within its time-slice (e.g., a Wilson loop or a single plaquette). With $|A_i| = O(1)$ and temporal distance t in lattice units, we obtain exponential decay at rate $1/\xi = \Omega(\kappa_*)$. Since $\|\mathcal{O}_i\|_\infty \ge \|\mathcal{O}_i\|_{L^2}$ for any probability measure, the L^2-formulation reads:

$$\left|\langle \mathcal{O}_1(0) \, \mathcal{O}_2(t) \rangle^{\mathrm{conn}}_{\mu_\Lambda}\right| \le C \, e^{-\kappa_* \, t} \, \|\mathcal{O}_1\|_{L^2} \, \|\mathcal{O}_2\|_{L^2},$$

with a constant C independent of $|\Lambda|$. One may take

$$C = C_1 |A_1||A_2| \frac{\|\mathcal{O}_1\|_\infty}{\|\mathcal{O}_1\|_{L^2}} \frac{\|\mathcal{O}_2\|_\infty}{\|\mathcal{O}_2\|_{L^2}},$$

which is $O(1)$ for local observables with $|A_i| = O(1)$. This exponential decay in Euclidean time translates into a spectral gap for the normalised Markov operator P. The operator satisfies $\lambda_0(P) = 1$ and corresponds to the transfer-operator eigenvalue ratio $\lambda_1(\hat{T})/\lambda_0(\hat{T})$. Writing $\langle \mathcal{O}_1(0) \, \mathcal{O}_2(t) \rangle^{\mathrm{conn}} = \langle f, P^t g \rangle_{L^2(\pi)}$ for mean-subtracted observables, the exponential bound implies $\|P^t f\|_{L^2(\pi)} \le C' e^{-\kappa_* t} \|f\|_{L^2(\pi)}$. Hence $\lambda_1(P) \le \inf_t (C')^{1/t} e^{-\kappa_*} = e^{-\kappa_*} < 1$. This bound extends from bounded local gauge-invariant functions to all of $L^2(\pi)$ by density and by the contraction property of P.

The lattice mass gap is therefore

$$\Delta_\Lambda = -\log \lambda_1(P) = \log \frac{\lambda_0(\hat{T})}{\lambda_1(\hat{T})} \ge \kappa_* > 0, \tag{25}$$

where $\kappa_* = (1 - \eta(\beta)) \, e^{-2 \, d_\ell \, \beta \, C_\rho}$ is the Poincaré constant from Step 2. This establishes exponential clustering with a volume-independent rate in the strong-coupling regime. $\square$

Corollary 4.4 (Explicit bound for $SU(2)$ in $d = 4$)**.** *For $G = SU(2)$ in $d = 4$ dimensions with the standard Wilson action in the fundamental representation and coupling $\beta < 2 \, \mathrm{artanh}(1/6) \approx 0.336$, the lattice mass gap satisfies*

$$\Delta_\Lambda \ge \kappa_* = \Big(1 - 6 \, \tanh(\beta/2)\Big) \, e^{-12\beta} > 0. \tag{26}$$

In particular, the bound is independent of the lattice volume $|\Lambda|$.

Proof. We evaluate each ingredient of Theorem 4.1 explicitly.

Step 1 (Haar spectral gap). For the heat-bath (Glauber) dynamics on a compact Lie group G with Haar measure, the Poincaré constant is $\kappa_{\mathrm{Haar}}(G) = 1$, since a single complete resample from the Haar measure reduces the conditional variance to zero. (Note: the spectral gap of the Laplace–Beltrami operator on $SU(2) \cong S^3$ with the round metric is $\lambda_1 = 3$, corresponding to the $l = 1$ spherical harmonic; this would be relevant if the Dirichlet form were the gradient form $\int |\nabla f|^2 \, d\mu$ rather than the heat-bath form $\sum_\ell \mathrm{Var}_\ell(f)$ used here.)

Step 1 (Holley–Stroock). In $d = 4$, each link belongs to $d_\ell = 2(d-1) = 6$ plaquettes. For $SU(2)$ in the fundamental representation ($N = 2$), $|\mathrm{Re}\,\mathrm{tr}(A_p U)| \le 2$, so the oscillation of the conditional potential satisfies $\mathrm{osc}(W) \le 2 \cdot 6 \cdot \beta = 12\beta$. By the Holley–Stroock lemma:

$$\kappa_{\text{link}} \ge 1 \cdot e^{-12\beta} = e^{-12\beta}.$$

Step 2 (Dobrushin constant via tanh*-bound).* For $SU(2)$ in the fundamental representation ($C(G) = 1$), the tanh-bound from Theorem 4.1, Step 2 gives $c_{\ell,\ell'} \le n_{\ell,\ell'} \tanh(\beta/2)$ for each neighbouring pair. Each link ℓ belongs to $d_\ell = 6$ plaquettes, so the Dobrushin constant is:

$$\eta(\beta) \le 6 \cdot \tanh(\beta/2).$$

The Dobrushin uniqueness condition $\eta(\beta) < 1$ holds for $\beta < \beta_0 = 2\,\mathrm{artanh}(1/6) \approx 0.336$.

Combining. By the Zegarlinski criterion (Step 2 of Theorem 4.1),

$$\Delta_\Lambda \ge \kappa_* = \Big(1 - 6\,\tanh(\beta/2)\Big)\, e^{-12\beta} > 0 \quad \text{for } \beta < 2\,\mathrm{artanh}(1/6).$$

For example, at $\beta = 0.20$: $\kappa_* = (1 - 6 \cdot 0.0997) \cdot e^{-2.4} \approx 0.402 \cdot 0.0907 \approx 0.0365$. □

Proposition 4.5 (Defective Dobrushin with typical-set improvement)**.** *For each conditional single-link measure $\mu(\cdot \mid \eta)$ on $G = \mathrm{SU}(N)$, let $S(\eta) \subset G$ denote the typical set (Equation 30). Suppose:*

(DD1) ***Improved bound on typical set:*** $C_P(\mu(\cdot \mid \eta)|_{S(\eta)}) \le C_{\text{typ}}(\eta)$ *with* $C_{\text{typ}} \le e^{-c\sqrt{\beta}}$.

(DD2) ***Small defect:*** $\mu(S(\eta)^c \mid \eta) \le \delta(\eta)$ *with* $\delta(\eta) \le e^{-c'\beta}$.

Then the conditional measure satisfies a defective Poincaré inequality*:*

$$\mathrm{Var}_{\mu(\cdot|\eta)}(f) \;\le\; C_{\text{typ}}(\eta) \cdot \mathcal{E}(f) \;+\; \delta(\eta) \cdot \|f\|_\infty^2. \tag{27}$$

Proof sketch. Decompose: $\mathrm{Var}(f) = \mathrm{Var}(f \cdot \mathbf{1}_S) + \mathrm{Var}(f \cdot \mathbf{1}_{S^c}) +$ cross terms. The first term is controlled by (DD1); the second and cross terms are bounded by $\delta \cdot \|f\|_\infty^2$ using (DD2) and Cauchy–Schwarz. □

Remark 4.6 (Annealed Dobrushin from defective Poincaré)*.* To lift Proposition 4.5 to the full lattice measure, define *annealed* influence coefficients

$$\tilde{c}_{ij} := \mathbb{E}_\eta\Big[c_{ij}(\eta)\Big], \tag{28}$$

where the expectation is over the boundary configuration η drawn from the lattice Gibbs measure. If the annealed Dobrushin matrix $\tilde{C} = (\tilde{c}_{ij})$ satisfies $\|\tilde{C}\|_\infty < 1$, then the lattice measure satisfies a Poincaré inequality with constant

$$C_P^{\text{lattice}} \;\le\; \frac{\sup_\eta C_{\text{typ}}(\eta) + \sup_\eta \delta(\eta) \cdot \mathrm{diam}(G)^2}{1 - \|\tilde{C}\|_\infty}. \tag{29}$$

Since $C_{\text{typ}} \le e^{-c\sqrt{\beta}}$ and $\delta \le e^{-c'\beta}$, the defect is exponentially suppressed and the lattice Poincaré constant inherits the typical-set improvement $e^{-O(\sqrt{\beta})}$ rather than the naive $e^{-O(\beta)}$.

This extends the validity of Theorem 4.1 to larger β: the Dobrushin condition $\|\tilde{C}\|_\infty < 1$ is weaker than the pointwise condition $\|C\|_\infty < 1$ used in Step 2, since averaging over η reduces the worst-case influence. The precise improvement in β_0 requires numerical evaluation of $\tilde{c}_{ij}$ (see Open Directions).

Remark 4.7 (Numerical Dobrushin coefficients for SU(2)). Monte Carlo simulation of 2D SU(2) lattice gauge theory (heatbath updates, $L = 4$) confirms the defective regime. The annealed Dobrushin coefficient $\eta = \max_i \sum_{j\sim i} \tilde{c}_{ij}$ satisfies $\eta \gg 1$ for all tested β values ($\eta \approx 46$ at $\beta = 1$, growing to $\eta \approx 76$ at $\beta = 8$). This confirms that the standard Dobrushin–Shlosman criterion fails and the hierarchical RG approach of Section 5.9 is necessary. The monotonic growth $\eta(\beta)$ is consistent with the expected polynomial degradation $\kappa_k \geq \kappa_0(1 - C/k^2)$ from Proposition 5.19. Script: `compute_dobrushin_su2.py` (to be made available as supplementary material upon publication).

Remark 4.8 (Birkhoff contraction along the RG cascade). The Birkhoff projective contraction coefficient $\tau_B(R_k)$ was computed for the SU(2) transfer matrix across $K = 6$ RG levels and $\beta \in \{1, 2, 4, 6, 8, 10, 15, 20\}$. For $\beta \leq 6$, $\tau_B(R_k) < 1$ *uniformly* across all scales, consistent with Proposition 5.19 in this toy discretisation. For $\beta \geq 8$, individual $\tau_B(R_k)$ approach 1 at the coarsest scale, but the Kingman–Lyapunov diagnostic $\lambda = \langle \log \tau_B \rangle$ remains negative on the tested grid ($\lambda \approx -0.18$ at $\beta = 8$, $\lambda \approx -0.04$ at $\beta = 10$). This supports the search for a summable-defect RG criterion, but it does not by itself prove mass-gap persistence without a uniform summability or gap-transport estimate. Script: `compute_birkhoff_rg.py` (to be made available as supplementary material upon publication).

Proposition 4.9 (Restricted Poincaré inequality for a single link on the typical set)**.** *Let G be a compact simple Lie group, ℓ a link on the hypercubic lattice in $d = 4$ dimensions, and $\mu_\ell(\cdot \mid \eta) \propto e^{-W(U_\ell;\eta)}\, dU_\ell$ the single-link conditional measure for boundary configuration η. Denote by*

$$\overline{W}(\eta) := \int_G W(U;\eta)\, d\mu_\ell(U \mid \eta)$$

the expected potential under μ_ℓ. For $\varepsilon > 0$, define the typical set

$$T_\varepsilon(\eta) := \Big\{U \in G : \Big|W(U;\eta) - \overline{W}(\eta)\Big| \leq \varepsilon\Big\}. \tag{30}$$

Let $\mu_\ell^{T_\varepsilon}$ denote the conditional measure μ_ℓ restricted to $T_\varepsilon(\eta)$.

Part A (Gradient-form Poincaré for the full single-link measure). *If the Dirichlet form is the gradient form $\mathcal{E}_\nabla(f) = \int_G |\nabla_G f|^2\, d\mu_\ell$ (where ∇_G is the gradient with respect to the bi-invariant metric on G), the Holley–Stroock bounded perturbation lemma [14] gives:*

$$\mathrm{Var}_{\mu_\ell}(f) \leq \frac{1}{\kappa_\nabla} \int_G |\nabla_G f|^2\, d\mu_\ell, \qquad \kappa_\nabla \geq \lambda_1^{\mathrm{LB}}(G) \cdot e^{-\mathrm{osc}(W)}, \tag{31}$$

where $\lambda_1^{\mathrm{LB}}(G)$ is the first nonzero eigenvalue of the Laplace–Beltrami operator on G with the bi-invariant metric normalized so that $\mathrm{vol}(G) = 1$.

For $G = SU(2) \cong S^3$ (round metric, unit volume): $\lambda_1^{\mathrm{LB}} = 4\pi^2 \cdot 3/\mathrm{vol}(S^3)$; with the standard normalization $\mathrm{vol}(G) = 1$, the eigenvalue is $\lambda_1 = 3$ (the $l = 1$ spherical harmonic on S^3). In $d = 4$, $\mathrm{osc}(W) \leq 12\beta$ (Corollary 4.4), giving:

$$\kappa_\nabla^{SU(2)} \geq 3\, e^{-12\beta}. \tag{32}$$

Proof of Part A. The Haar measure on G (normalised to unit volume) satisfies the Poincaré inequality

$$\mathrm{Var}_{\mathrm{Haar}}(f) \le \frac{1}{\lambda_1^{\mathrm{LB}}(G)} \int_G |\nabla_G f|^2 \, d\mathrm{Haar}$$

by the Lichnerowicz–Obata theorem (or direct computation for S^3). The conditional measure $d\mu_\ell = e^{-W}/Z \, d\mathrm{Haar}$ is a bounded perturbation of the Haar measure with density ratio bounded by $e^{\mathrm{osc}(W)}$. By the Holley–Stroock lemma [14], any log-concave perturbation of a measure satisfying a Poincaré inequality with constant κ_0 again satisfies a Poincaré inequality with constant $\kappa_0 \cdot e^{-\mathrm{osc}(W)}$. More precisely: for any probability measure μ with $d\mu = e^{-\phi} \, d\nu/Z$ and $\mathrm{osc}(\phi) = \sup \phi - \inf \phi < \infty$, if ν satisfies $\mathrm{Var}_\nu(f) \le \kappa_0^{-1} \, \mathcal{E}_\nu(f)$, then μ satisfies $\mathrm{Var}_\mu(f) \le (\kappa_0 \, e^{-\mathrm{osc}(\phi)})^{-1} \, \mathcal{E}_\mu(f)$. Applying this with $\nu = $ Haar, $\phi = W$, and $\kappa_0 = \lambda_1^{\mathrm{LB}}(G)$ yields (31).

For $SU(2) \cong S^3$: the spectrum of the Laplace–Beltrami operator on S^3 with the round metric (normalised to vol $= 1$) has eigenvalues $\lambda_l = l(l+2)$ for $l = 0, 1, 2, \ldots$, so $\lambda_1 = 1 \cdot 3 = 3$. (Under the standard physicist's normalisation with $\mathrm{vol}(S^3) = 2\pi^2$, the eigenvalues are $l(l+2)/R^2$; normalising to unit volume rescales but preserves $\lambda_1 = 3$.) Combined with $\mathrm{osc}(W) \le 12\beta$, this gives (32). □

Part B (Improved Poincaré constant on the typical set T_ε). *For any $\varepsilon > 0$ with $\mu_\ell(T_\varepsilon) > 0$, the restricted measure $\mu_\ell^{T_\varepsilon}$ satisfies:*

$$\mathrm{Var}_{\mu_\ell^{T_\varepsilon}}(f) \le \frac{1}{\kappa_\varepsilon} \int_{T_\varepsilon} |\nabla_G f|^2 \, d\mu_\ell^{T_\varepsilon}, \tag{33}$$

with

$$\kappa_\varepsilon \ge \lambda_1^{\mathrm{LB}}(G) \cdot e^{-2\varepsilon} \cdot \mu_\ell(T_\varepsilon)^2. \tag{34}$$

In particular, for $\varepsilon = c\sqrt{\beta}$ with a constant $c > 0$ chosen such that $\mu_\ell(T_\varepsilon) \ge 1/2$ (which is guaranteed for small enough c by concentration of the Wilson action—see the proof), the restricted Poincaré constant satisfies

$$\kappa_\varepsilon \ge \frac{\lambda_1^{\mathrm{LB}}(G)}{4} \cdot e^{-2c\sqrt{\beta}}, \tag{35}$$

which degenerates only as $e^{-O(\sqrt{\beta})}$ instead of $e^{-O(\beta)}$—a qualitative improvement over the unrestricted bound (31) for large β.

Proof of Part B. The proof proceeds in three steps.

Step 1 (Oscillation reduction on the typical set). On $T_\varepsilon(\eta)$, the potential satisfies $W(U) \in [\overline{W} - \varepsilon, \overline{W} + \varepsilon]$ by definition, so

$$\mathrm{osc}_{T_\varepsilon}(W) := \sup_{T_\varepsilon} W - \inf_{T_\varepsilon} W \le 2\varepsilon.$$

This is the key improvement: the full oscillation $\mathrm{osc}(W) \le 2 \, d_\ell \, \beta$ (which grows linearly in β) is replaced by 2ε, which can be chosen to grow much more slowly.

Step 2 (Restriction penalty). The restricted measure $\mu_\ell^{T_\varepsilon} = \mu_\ell(\cdot \mid T_\varepsilon)$ has density $d\mu_\ell^{T_\varepsilon}/d\mathrm{Haar}|_{T_\varepsilon} = e^{-W}/(\mu_\ell(T_\varepsilon) \cdot Z)$ on T_ε. By the localisation lemma (see Milman [24], or the elementary argument below), restricting a Poincaré inequality from a set of full measure to a subset A with $\mu(A) > 0$ incurs a penalty factor of at most $1/\mu(A)^2$. Specifically: if μ

satisfies $\operatorname{Var}_\mu(f) \leq C_P\, \mathcal{E}_\mu(f)$, then for any measurable A with $\mu(A) \geq p > 0$, the restricted measure $\mu_A = \mu(\cdot \mid A)$ satisfies

$$\operatorname{Var}_{\mu_A}(f) \leq \frac{C_P}{p^2}\, \mathcal{E}_{\mu_A}(f).$$

This follows from $\operatorname{Var}_{\mu_A}(f) \leq \frac{1}{p} \operatorname{Var}_\mu(f \cdot \mathbf{1}_A)/\mu(A) \leq \frac{1}{p^2} \operatorname{Var}_\mu(\tilde{f})$ where $\tilde{f}$ is the extension by zero, combined with the monotonicity of the Dirichlet form.

Step 3 (Combining). The restricted Haar measure $\mathrm{Haar}|_{T_\varepsilon}$ satisfies a Poincaré inequality by the localisation argument of Step 2: since $\mathrm{Haar}(T_\varepsilon) \geq 3/4$ (by the concentration bound below), restricting the full Haar–Poincaré inequality with constant $\lambda_1^{\mathrm{LB}}(G)$ to T_ε incurs a penalty of at most $1/\mathrm{Haar}(T_\varepsilon)^2 \leq 16/9$, giving a restricted Haar–Poincaré constant $\geq (9/16)\, \lambda_1^{\mathrm{LB}}(G)$. The restricted conditional measure $\mu_\ell^{T_\varepsilon}$ is then a bounded perturbation of this restricted Haar measure with oscillation at most 2ε (Step 1). By Holley–Stroock on the restricted domain:

$$\kappa_\varepsilon \geq \lambda_1^{\mathrm{LB}}(G) \cdot e^{-2\varepsilon} \cdot \mu_\ell(T_\varepsilon)^2.$$

Concentration bound for $\mu_\ell(T_\varepsilon)$. The potential $W(U;\eta) = -(\beta/N) \sum_{p \ni \ell} \operatorname{Re}\operatorname{tr}(A_p U)$ is a sum of $d_\ell = 6$ bounded terms, each with $|\operatorname{Re}\operatorname{tr}(A_p U)| \leq N$. By the measure concentration inequality for Lipschitz functions on compact groups (see e.g. Ledoux [Ledoux(2001)], Chapter 2), or directly by Chebyshev's inequality applied to $\operatorname{Var}_{\mu_\ell}(W) \leq \beta^2 \cdot C(d, G)$ (which follows from the bounded oscillation), we obtain:

$$\mu_\ell\Big(\big|W - \overline{W}\big| > \varepsilon\Big) \leq \frac{\operatorname{Var}_{\mu_\ell}(W)}{\varepsilon^2} \leq \frac{C \cdot \beta^2}{\varepsilon^2}.$$

Choosing $\varepsilon = c\sqrt{\beta}$ with $c = \sqrt{4C}$ gives $\mu_\ell(T_\varepsilon) \geq 1 - 1/4 = 3/4 > 1/2$, hence $\mu_\ell(T_\varepsilon)^2 \geq 1/4$.

Remark on sub-Gaussian improvement: The Chebyshev bound used above is not optimal. Since $W(U;\eta)$ is a Lipschitz function on the compact Riemannian manifold G (with positive Ricci curvature), the Lévy–Milman concentration inequality (Ledoux [Ledoux(2001)], Theorem 2.3) yields the stronger sub-Gaussian bound $\mu_\ell(|W - \overline{W}| > \varepsilon) \leq C\, e^{-c\varepsilon^2/\beta^2}$, which would improve the restricted Poincaré constant to $e^{-O(\beta\sqrt{\log \beta})}$. We use the weaker Chebyshev bound here for simplicity; the sub-Gaussian improvement is noted as a potential enhancement. □

Remark 4.10 (Significance of the restricted Poincaré bound)*.* The restricted Poincaré inequality (Proposition 4.9, Part B) demonstrates that the exponential degeneration of the Holley–Stroock constant in β is partly an artifact of rare large-deviation configurations. On the typical set, the Poincaré constant degenerates only as $e^{-O(\sqrt{\beta})}$ instead of $e^{-O(\beta)}$. This suggests a path towards β-independent gap bounds:

1. If the typical-set restriction can be combined with the Dobrushin–Zegarlinski machinery (Step 2 of Theorem 4.1), replacing the full single-link Poincaré constant by the restricted one, the Dobrushin threshold would improve from $\beta_0 = O(1)$ to $\beta_0' = O(1)$ with a much larger prefactor.

2. For the continuum limit, the $e^{-O(\sqrt{\beta})}$ degeneration combined with $\beta(a) \sim -\frac{1}{b_0} \log(a\Lambda_{\mathrm{QCD}})$ (asymptotic freedom) gives $\kappa_\varepsilon \sim \exp(-O(\sqrt{\log(1/a)}))$, which tends to zero *much more slowly* than $e^{-O(\beta)} \sim (a\Lambda_{\mathrm{QCD}})^{O(1/b_0)}$.

This is listed as Strategy 2 in the Roadmap (Section 5.4) and, if combined with a suitable multi-scale decomposition, could potentially yield the physical mass gap without requiring Kirk's complete analyticity. Our main results (Theorems 3.4 and 4.2) do not depend on Kirk's work; his results are cited as evidence for the plausibility of CL1–CL3.

Lemma 4.11 (OS reconstruction: clustering implies mass gap)**.** *Assume the following:*

(CL1) ***OS-positive continuum limit.*** *There exists a sequence $(a_n, \Lambda_n, \beta_n)$ with $a_n \to 0$, $\Lambda_n \uparrow \mathbb{Z}^4$, and $\beta_n = \beta(a_n)$ tuned according to asymptotic freedom [26, 27], such that the renormalized Schwinger functions $S^{(k)}_{a_n,\Lambda_n}$ converge for a dense class of gauge-invariant local observables and satisfy the Osterwalder–Schrader axioms.*

(CL2) ***Uniform exponential clustering in physical distance.*** *There exist constants $A, \Delta_0 > 0$ such that for all n and all gauge-invariant observables $\mathcal{O}_1, \mathcal{O}_2$ in the above class,*

$$\left|\langle \mathcal{O}_1(x)\, \mathcal{O}_2(0)\rangle^{\mathrm{conn}}_{a_n,\Lambda_n}\right| \le A\, e^{-\Delta_0\, |x|_{\mathrm{phys}}}, \tag{36}$$

where $|x|_{\mathrm{phys}} = a_n\, |x|_{\mathrm{lat}}$ is the physical distance.

Remark 4.12 (Relaxation of CL2)*.* Assumption CL2 (uniform log-Sobolev inequality) can be replaced by the structurally weaker condition:

CL2'. *Uniform Poincaré inequality in physical distance scale:* There exists $C > 0$ such that for all lattice spacings $a > 0$,

$$\mathrm{Var}_{\mu_\Lambda}(f) \le C \cdot \xi(a)^2 \cdot \mathcal{E}_\Lambda(f, f)$$

where $\xi(a)$ is the correlation length and $\mathcal{E}_\Lambda$ is the Dirichlet form. This is equivalent to requiring that the mass gap in physical units, $m_{\mathrm{phys}} = 1/\xi(a)$, remains bounded below: $m_{\mathrm{phys}} \ge m_0 > 0$.

CL2' is strictly weaker than CL2 because it allows the lattice Poincaré constant $\kappa(a)$ to degrade as $\kappa(a) \sim a^2/\xi(a)^2$, provided $\xi(a)$ grows at most polynomially in $1/a$. This is consistent with asymptotic freedom, where $\xi(a) \sim a^{-1} e^{-c/g(a)^2}$ with $g(a) \to 0$ logarithmically.

(CL3) ***Uniform normalisation.*** *The renormalised observables satisfy $\|\mathcal{O}_i\|_{L^2(\mu_{a_n,\Lambda_n})} \le C$ uniformly in n, so that A in (36) does not diverge.*

Then the reconstructed quantum field theory on $\mathbb{R}^4$ possesses a mass gap $\Delta \ge \Delta_0 > 0$.

Caveat: Assumption (CL2) is essentially equivalent to the mass gap itself, since it postulates uniform exponential decay of connected correlators in physical units. The non-trivial content of this lemma is therefore *not* an independent derivation of the mass gap, but rather the systematic reduction of the continuum problem to three independently attackable conditions (CL1)–(CL3), of which (CL2) carries the bulk of the difficulty.

Proof. **Step 1 (Pointwise limit of correlators).** Fix gauge-invariant local observables $\mathcal{O}_1, \mathcal{O}_2$ in the dense class of (CL1). By (CL3), the connected two-point functions satisfy $|C_n(t)| \le \|\mathcal{O}_1\|_{L^2}\, \|\mathcal{O}_2\|_{L^2} \le C^2$ for all n. Combined with (CL2), $|C_n(t)| \le \min(C^2,\, A\, e^{-\Delta_0\, t}) =: g(t)$. Since $g \in L^1(\mathbb{R}_+)$, the dominated convergence theorem applies, and the pointwise limit $C(t) := \lim_n C_n(t)$ (which exists by (CL1)) satisfies $|C(t)| \le A\, e^{-\Delta_0\, t}$ for all $t > 0$.

Step 2 (Spectral reconstruction). By (CL1), the limiting Schwinger functions satisfy the OS axioms. The OS reconstruction theorem [3] produces a Hilbert space $\mathcal{H}$, a positive self-adjoint Hamiltonian H, and a vacuum Ω with $H\Omega = 0$. For any state $\Psi = \mathcal{O}\,\Omega \in \mathcal{H} \ominus \mathbb{C}\Omega$:

$$\langle \Psi,\, e^{-tH}\,\Psi\rangle = C(t) \le A\, e^{-\Delta_0\, t}.$$

By the spectral theorem, $\langle \Psi,\, E_{[0,\Delta_0)}(H)\,\Psi\rangle = 0$ for all such Ψ. Since the $\mathcal{O}\,\Omega$ span a dense subspace of $\mathcal{H} \ominus \mathbb{C}\Omega$ (by the Reeh–Schlieder property, which follows from (CL1)), we conclude $E_{[0,\Delta_0)}(H) = 0$ on $\mathcal{H} \ominus \mathbb{C}\Omega$, hence $\inf \sigma(H|_{\mathcal{H}\ominus\mathbb{C}\Omega}) \ge \Delta_0 > 0$.

Role of (CL3)*:* The uniform normalisation assumption ensures that the majorant $g(t)$ is independent of n, making the dominated convergence theorem applicable. Without (CL3), the amplitude A could diverge with n, invalidating the lattice-to-continuum passage. □

Remark 4.13 (Relation to the Millennium Problem and the role of Lemma 4.11). Lemma 4.11 is best understood as a *reconstruction lemma*: it shows that the OS reconstruction machinery faithfully translates Euclidean exponential clustering into a Hilbert-space mass gap. Assumption (CL2), however, is essentially equivalent to the mass gap itself, since it postulates uniform exponential decay of connected correlators in physical units. Thus, Lemma 4.11 does *not* provide an independent derivation of the mass gap; rather, it reduces the continuum problem to verifying (CL1)–(CL3).

The genuine content of this paper lies in Theorem 4.1: the lattice theory has a gap $\Delta_\Lambda \ge \kappa_* > 0$ in *lattice* units for $\beta < \beta_0$ (strong coupling), uniformly in the volume $|\Lambda|$. However, in the continuum limit $a \to 0$, the lattice gap Δ_{lat} shrinks to zero while the physical gap $\Delta_{\mathrm{phys}} = \Delta_{\mathrm{lat}}/a$ should remain positive. Proving this requires non-perturbative control over the gap-scaling along the renormalisation group trajectory $\beta(a)$, which is the core of the Millennium Problem.

Balaban's renormalisation group analysis [7, 8, 9] provides UV stability of the effective action across scales—a necessary ingredient but not a complete proof of the continuum limit with OS axioms and mass gap in 4D.

Corollary 4.14 (Mass gap for $SU(N)$)**.** *For $G = SU(N)$ with $N \ge 2$, if assumptions (CL1)–(CL3) are satisfied, the quantum Yang–Mills theory on $\mathbb{R}^4$ possesses a mass gap $\Delta > 0$.*

Theorem 4.15 (Conditional mass gap)**.** *Let G be a compact simple Lie group and let $\mu_{\Lambda,\beta}$ denote the lattice Yang–Mills measure on a finite lattice $\Lambda \subset a\mathbb{Z}^4$ at coupling β. Assume:*

(T1) Tail bound: *There exists $C, \alpha > 0$ such that $\mu_{\Lambda,\beta}(\|U_l - \mathbf{1}\| > t) \le Ce^{-\alpha t^2 \beta}$ uniformly in Λ and $l \in \Lambda$.*

(T2) Gap transport: *The Poincaré constant satisfies $\kappa(2a) \ge c \cdot \kappa(a)$ for a universal $c > 0$ (Conjecture 5.14).*

(T3) Reflection symmetry: *The block-spin kernel commutes with lattice reflections.*

Then the continuum Yang–Mills theory on $\mathbb{R}^4$ (constructed via the thermodynamic and continuum limits) has a mass gap $m > 0$, satisfies the Osterwalder–Schrader axioms, and the vacuum is unique.

Proof sketch. Under (T1), the single-link measure is sub-Gaussian, which implies the Typical-Set Poincaré bound (Proposition 4.9) with improved scaling $e^{-O(\sqrt{\beta})}$. Under (T2), this bound propagates through the RG hierarchy (Proposition 5.17), yielding a uniform mass gap in physical units. Under (T3), the RP preservation (Proposition 3.4) ensures the Osterwalder–Schrader reconstruction theorem applies. □

4.1 Post-Zookeeper transfer: lattice gap and area law as the main path

The post-Zookeeper reading changes the emphasis of this paper. The Shenfeld/SU(N) extension remains a useful stochastic-continuum bridge, but the main route should now be organised around the lattice results on mass gap, Poincare/log-Sobolev inequalities, Wilson-loop area law, and the $'t$ Hooft regime.

The updated CORE status is deliberately more modest for physics: Zookeeper closes the Riemann $C^{(2)}$/MS2 step in the CCM/Fourier branch, while RFEP v1.8 imports only the normal-form checklist below. It does not by itself prove the Yang–Mills continuum mass gap; Prime-Hub contributes only a boundary/defect bridge, not a solved physical operator.

The transfer slots are:

Operator/form. The transfer operator, heat-bath/Glauber generator, and Wilson-loop observables.

Defect. Failure of uniform correlation decay, RG-transport defect, and continuum reconstruction defect.

Approximation. Finite lattices, slab sigma-models, $'t$ Hooft large-N scaling, and block-spin RG.

Gap. Poincare/LSI/Bakry–Emery constants, exponential decay of correlations, and the area-law exponent.

Limit. Finite volume $\to$ infinite volume $\to$ large N or $'t$ Hooft regime $\to$ continuum/OS reconstruction.

Thus the proof programme should be stated as

$$\begin{aligned}&\text{finite-volume Poincare/LSI}\\ &\implies \text{infinite-volume lattice mass gap}\\ &\implies \text{Wilson area law}\\ &\implies \text{conditional continuum transfer.}\end{aligned}$$

The U(N) to SU(N) decomposition is especially useful here: the U(1) field can be treated as an environment, while the SU(N) marginal carries the non-abelian gap. After this reorganisation the single red block is the continuum-scaling statement that a positive lattice gap transports to a positive physical mass gap.

Proof status summary

5 Discussion

5.1 Relation to Lattice QCD Results

The variational argument presented here is consistent with extensive numerical evidence from lattice computations. For $SU(3)$, lattice simulations yield a mass gap (identified with the lightest glueball mass) of approximately $\Delta \approx 1.5$–1.7 GeV [11, 12]. Our approach

Table 1: Proof status summary

Component	Status	Reference
Proven (unconditional):		
Holley–Stroock PI ($\beta < \beta_0$)	Theorem	Thm. 4.1, Step 1
Reflection positivity (all β)	Lemma	Lem. 3.1
Dobrushin–Zegarlinski ($\beta < \beta_0$)	Theorem	Thm. 4.1, Step 2
Conditional (on explicit assumptions):		
Mass gap (CL1+CL2'+CL3)	Conditional Lemma	Lem. 4.11, Thm. 4.15
Polynomial LSI scaling	Conditional Prop.	Prop. 5.19
Kingman/summable-defect route	Conditional criterion	Cor. 5.22
Numerically confirmed:		
Defective Dobrushin ($\eta \gg 1$)	Monte Carlo	Rem. 4.7
Birkhoff $\tau_B < 1$ ($\beta \leq 6$)	Transfer matrix	Rem. 4.8
Kingman $\lambda < 0$ (tested β grid)	Transfer matrix diagnostic	Rem. 4.8
Open:		
Summable RG defect analytically for $\beta \to \infty$	Conjecture	Sec. 5.11
Gap transport (all β)	Conjecture	Conj. 5.14

does not predict the numerical value of Δ; it proves strict positivity in the strong-coupling lattice regime and formulates the additional transport conditions needed beyond that regime.

To put the strong-coupling threshold in perspective: for $SU(2)$ in $d = 4$, the tanh-based Dobrushin condition requires $\beta < \beta_0 = 2 \operatorname{artanh}(1/6) \approx 0.336$. Lattice Monte Carlo studies of $SU(2)$ gauge theory typically operate at $\beta \approx 2$–3 (scaling regime), which is roughly one order of magnitude larger. The strong-coupling regime is thus far from the physically relevant continuum limit, as expected for any argument based on cluster expansions (which converge only at high temperature/strong coupling).

5.2 The Role of Strict Convexity

The strict convexity of $\mathcal{F}$ plays the role that coercivity of the Hamiltonian plays in conventional approaches. The key insight is that massless excitations—if they existed—would correspond to flat directions in $\mathcal{F}$, i.e., perturbations $\delta\mu$ with $\delta^2\mathcal{F}[\mu_\Lambda](\delta\mu, \delta\mu) = 0$. But the entropy term in $\mathcal{F}$ ensures that no such flat directions exist among gauge-invariant perturbations.

This mechanism is analogous to the generation of a mass gap in the $(\nabla\varphi)^4$ scalar field theory by the entropy of fluctuations, where the perturbative masslessness is an artifact of ignoring the full non-perturbative measure.

Remark 5.1 (Formal FST convexity and transport caveat). Formally, the KL part of $\mathcal{F}[\mu]$ suggests a Wasserstein-convexity mechanism with scale comparable to $1/\beta$. Turning this into a Talagrand or Poincaré estimate for the gauge-field measure would require a verified geodesic convexity/curvature-dimension statement on the relevant gauge orbit or slice, plus compatibility with reflection positivity. We therefore use this remark only as a transport heuristic; it is not an additional proof of a linear lower bound $\kappa \gtrsim 1/\beta$.

Remark 5.2 (Annealed influence as a universal gap mechanism). The Dobrushin–Zegarlinski machinery used in this paper—bounding the spectral radius of the influence matrix to establish volume-independent gaps—is not specific to lattice gauge theory. The same mechanism applies, *mutatis mutandis*, to two other instantiations in the FST programme:

1. *Turbulence (shell cascades):* The shell-by-shell energy transfer defines an influence matrix C_{jk} on the inertial range. The Dobrushin uniqueness condition $\rho(C) < 1$ is equivalent to the stability of K41 as the unique Gibbs state on the cascade lattice (Remark 4.22 of [Geiger(2026d)]). The "defective Dobrushin" extension handles intermittency corrections.

2. *P vs NP (witness search):* For NP-complete instance families, the influence matrix on the witness-space Gibbs measure is conjectured to have $\rho(C) \geq 1$, obstructing rapid mixing and hence polynomial-time witness extraction (cf. the companion P vs NP paper [Geiger(2026e)]).

The *defective Poincaré/LSI lemma* thus functions as a universal template: either the influence is weak and a gap exists (YM strong coupling, TU inertial range), or the influence is strong and the gap is obstructed (YM weak coupling, P vs NP hard instances). The Pattern A architecture classifies these as two sides of the same coin.

Remark 5.3 (Deconfinement and finite temperature). For $SU(N)$ gauge theory at finite temperature in $d = 4$ (lattice with finite temporal extent N_τ), the deconfinement phase transition at $\beta_c(N_\tau)$ spontaneously breaks the $\mathbb{Z}(N)$ centre symmetry (the Polyakov loop acquires a nonzero expectation value). Above the critical temperature, the confining string tension vanishes and the spectrum changes qualitatively: the discrete glueball spectrum of the confining phase gives way to a continuum of scattering states, though massive screening modes (Debye mass $\sim gT$) persist. This transition is not relevant to the zero-temperature Millennium Problem (which concerns the infinite-volume, zero-temperature theory on $\mathbb{R}^4$), but it illustrates that the nature of the spectrum depends sensitively on the order of limits (spatial volume $\to \infty$ before temporal extent $\to \infty$).

5.3 Mixture decomposition approach

Remark 5.4 (Mixture decomposition for weak coupling)*.* For $\beta > \beta_0$, the global Holley–Stroock bound degrades because the oscillation $\mathrm{osc}(S) \to \infty$. A more refined approach decomposes the Gibbs measure into metastable components:

$$\mu = \sum_\alpha w_\alpha \mu_\alpha, \quad w_\alpha > 0, \quad \sum_\alpha w_\alpha = 1,$$

where each μ_α is concentrated near a local minimum of the Wilson action (a "well"). The spectral gap then satisfies

$$\Delta \geq \min\left(\min_\alpha \Delta_\alpha, \quad \Delta_{\mathrm{mix}}\right),$$

where Δ_α is the within-well Poincaré constant (computable via Bakry–Émery curvature on the orbit space $\mathcal{A}/\mathcal{G}$) and Δ_{mix} is the between-well mixing rate (controlled by the Eyring–Kramers formula for barrier crossing).

For SU(2) in 4D, the within-well curvature is bounded below by the Ricci curvature of S^3, giving $\Delta_\alpha \geq 3 - O(\beta^{-1})$. The between-well mixing rate depends on the instanton action: $\Delta_{\mathrm{mix}} \sim e^{-8\pi^2/g^2}$ (instanton tunnelling), which is exponentially small but strictly positive for any finite lattice.

5.4 Limitations and Open Problems

Our argument establishes the mass gap rigorously only in the strong-coupling regime ($\beta < \beta_0$) and conditionally for the continuum theory. The key open problems are:

1. **Gap-scaling under renormalisation:** The lattice gap $\Delta_\Lambda \geq \kappa_* > 0$ is proven for $\beta < \beta_0$ (Theorem 4.1). In the continuum limit, $\beta(a) \to \infty$ by asymptotic freedom, so the strong-coupling regime is eventually left. Whether the physical gap $\Delta_{\mathrm{phys}} = \Delta_{\mathrm{lat}}/a$ remains bounded below as $a \to 0$ is a non-perturbative question closely related to the structure of the renormalisation group flow. This is the core of the Millennium Problem.

   A structural argument for gap persistence would require establishing the absence of first-order phase transitions along the renormalisation flow (see Remark 6.2 for the broader structural context). This remains an open problem.

2. **Constructive existence of the continuum limit:** Assumptions (CL1)–(CL3) in Lemma 4.11 subsume both the existence of OS-positive Schwinger functions and uniform exponential clustering. Balaban's programme provides UV stability but does not close the full IR/constructive gap in 4D. *Note added (May 2026):* Kirk [18] has posted a Zenodo preprint on a conditional SU(2) continuum-limit construction

using Dobrushin–Shlosman complete analyticity and pro-polymer cluster expansion estimates. The preprint explicitly relies on weak-coupling hypotheses, and any companion zero-free-strip input remains to be independently verified. These results should therefore be treated as conditional preprint evidence, not as a closed proof of (CL1)–(CL3).

3. **Extension to all β:** The Dobrushin–Zegarlinski criterion (Step 2) yields $\kappa_* > 0$ only for $\beta < \beta_0$. While finite-volume analyticity precludes sharp phase transitions for fixed $|\Lambda|$, the persistence of a uniform gap for all $\beta > 0$ in the thermodynamic limit is an open conjecture (see Remark 4.2).

A complete resolution of the Millennium Problem requires establishing the constructive continuum limit *and* proving that the physical mass gap is preserved under the renormalisation group flow.

Remark 5.5 (Mass gap as RG fixed-point property). An alternative conceptual framework avoids the $\beta \to \infty$ limit entirely by reformulating the mass gap as a *fixed-point* statement. Define the RG-transformed free energy $\mathcal{F}_\ell[\mu] := \mathcal{F}[\mu; \beta(\ell), \Lambda(\ell)]$ at scale ℓ, where $\beta(\ell)$ and $\Lambda(\ell)$ follow the Wilson RG flow. The mass gap condition becomes: there exists a non-trivial RG fixed point $\mathcal{F}^*$ with spectral gap $\Delta^* > 0$, and the flow $\ell \mapsto \mathcal{F}_\ell$ converges to $\mathcal{F}^*$ in the infrared. In this formulation:

- The strong-coupling result (Theorem 4.1) establishes that $\mathcal{F}_0$ (UV starting point) has a gap.
- Asymptotic freedom guarantees the UV fixed point is Gaussian (free theory, no gap).
- The open question reduces to: does the RG flow from the Gaussian UV fixed point produce a confining IR regime with $\Delta^* > 0$? (Note: 4D Yang–Mills is believed to be confining rather than flowing to a non-trivial conformal fixed point; the "fixed point" here refers to a stable massive phase, not a conformal field theory.)

This shifts the problem from a limiting argument ($\beta \to \infty$, where the Holley–Stroock bounds degenerate) to a *stability analysis of the RG flow*—a question about the infrared behaviour of finitely many effective couplings, once the relevant truncation is identified. Establishing the absence of fixed-point bifurcations (no phase transitions) along the RG flow is a necessary condition for the existence of a unique IR fixed point; this remains an open problem.

Remark 5.6 (Mass gap as renormalisation-stable LSI property). Proposition 5.20 gives a renormalisation-stable log-Sobolev route: a family of LSI constants $\{\kappa_k\}_{k\geq 1}$ with summable Birkhoff defects, or an equivalent GT2 error budget, would transport the mass gap. The Kingman–Lyapunov exponent $\lambda = \langle \log \tau_B \rangle$ is a useful diagnostic, but without summability control it is not an equivalent characterisation. The relevant numerical quantity is still accessible (Remark 5.23: $\lambda \approx -0.18$ at $\beta = 8$, $\lambda \approx -4 \times 10^{-5}$ at $\beta = 20$) and structurally analogous to the Lyapunov exponent in dynamical systems theory.

Remark 5.7 (Analytical route to $\lambda < 0$: Doeblin minorisation on typical sets). The numerical observation $\lambda < 0$ on the tested β grid (Remark 5.23) admits a natural analytical proof strategy via the *Doeblin minorisation* technique for products of random positive operators.

Step 1: Random positive operator cocycle. On each RG level k, the block-spin transfer kernel R_k is a positive operator whose randomness derives from the boundary configuration under the Gibbs measure of the previous scale. This defines a positive operator cocycle $\{R_k\}_{k\geq 1}$.

Step 2: Typical-set Doeblin condition. The key analytical ingredient is a *minorisation on typical boundary configurations*: there exists an event $\mathcal{T}_k$ ("typical boundaries") with $\mathbb{P}(\mathcal{T}_k) \geq 1 - e^{-c\beta}$ and a fixed reference measure component ν such that on $\mathcal{T}_k$:

$$R_k(x, \cdot) \;\geq\; \alpha(\beta)\, \nu(\cdot) \qquad \text{(as measures)},$$

with $\alpha(\beta) > 0$ explicit. This is the Birkhoff-projective analogue of the defective Dobrushin condition used for the Poincaré inequality (Proposition 4.5).

Step 3: Minorisation $\Rightarrow$ negative drift. Doeblin's minorisation implies a quantitative Birkhoff contraction: $\tau_B(R_k) \leq 1 - \delta(\beta)$ on $\mathcal{T}_k$. The expected log-contraction satisfies:

$$\mathbb{E}[\log \tau_B(R_k)] \;\leq\; \mathbb{P}(\mathcal{T}_k) \cdot \log(1 - \delta(\beta)) \;+\; \mathbb{P}(\mathcal{T}_k^c) \cdot 0 \;\approx\; -(1 - e^{-c\beta})\, \delta(\beta) \;<\; 0,$$

since $\log \tau_B \leq 0$ always holds (positive operators contract in Hilbert's projective metric).

Step 4: Kingman's theorem. Two standard technical conditions remain. The boundary environment along the RG cascade must be stationary or ergodic (via the Markov property and translation symmetries), and $\log \tau_B(R_k)$ must be integrable. The latter follows from the tail bound $0 \geq \log \tau_B \geq -C \cdot \mathrm{osc}(W)$. Kingman's subadditive ergodic theorem then delivers $\lambda < 0$ as a pure drift result.

This strategy makes precise the numerical observation that individual $\tau_B(R_k)$ approach 1 at the coarsest scale while the averaged log-exponent remains negative—the signature of a *rare-bad / typical-good* mechanism analytically captured by minorisation plus Kingman. The programme reduces the analytical proof of $\lambda < 0$ to establishing the Doeblin constant $\alpha(\beta) > 0$ for the SU(2) block-spin kernel, which is a concrete problem in stochastic geometry on compact Lie groups.

For related analytical methods on products of random positive operators, see Bougerol–Lacroix [37].

5.5 Outlook

The variational free-energy approach may extend to Yang–Mills–Higgs theories and to the study of confinement, where the string tension can be interpreted as the Hessian of a suitably modified free-energy functional over Wilson loop observables.

5.6 Open Directions

The following remarks collect six conceptual threads that arise naturally from the variational framework developed above. They are intended as stimuli for future work rather than as proven results.

Remark 5.8 (Topological sectors as "no flat directions"). Every non-trivial topological deformation—i.e., every change of the instanton number $\nu = c_2(P) \in \mathbb{Z}$—requires crossing an energy barrier of order $8\pi^2/g^2$ (one-instanton action). This implies that the free energy $F[\mu]$ has *no flat directions* in the topological sector: any perturbation that changes the topology incurs a strictly positive cost. This provides a conceptual foundation for the positivity of the Hessian at the vacuum, complementing the quantitative Poincaré bound.

Remark 5.9 (Knotted flux tubes as candidates for the first massive excitation). Knotted chromoelectric flux tubes (glueball candidates) provide a physical interpretation of the first massive excitation above the vacuum. The mass gap m corresponds to the lightest glueball, whose stability is enforced by topological conservation: the flux tube cannot unwind without crossing a free-energy barrier. In the FST framework, this is the gap between the vacuum eigenvalue $\lambda_0 = 0$ of the transfer operator and the first excited eigenvalue $\lambda_1 = m$.

Remark 5.10 (Non-abelian Kramers–Wannier duality). A non-abelian Kramers–Wannier duality would map the weak-coupling regime ($\beta \to \infty$, $g \to 0$) to a dual strong-coupling description in terms of monopole/vortex variables. Under such a duality, the mass gap at weak coupling would correspond to confinement of dual monopoles at strong coupling—a well-understood regime. While no rigorous non-abelian KW duality exists, the 't Hooft electric–magnetic duality provides partial evidence, and the FST framework is agnostic to the choice of variables: the free energy $F[\mu]$ can in principle be reformulated in dual variables.

Remark 5.11 (Haar measure entropy: an entropic route to confinement). An alternative route to confinement: the Haar measure entropy $S_{\text{Haar}} = \log \text{vol}(G^{|\Lambda|})$ of the gauge group grows extensively with the lattice volume. Free (deconfined) quarks would require correlated configurations that occupy an exponentially smaller fraction of the gauge orbit space, with entropy deficit $\Delta S \sim L^3 \cdot \log N$. The RFEP predicts that the system selects the configuration maximising $\Phi = S - \beta E$ subject to stability constraints; for β below the deconfinement temperature, the entropic term dominates and forces colour-singlet (confined) states.

Remark 5.12 (RG-monotonicity candidate $M(\beta)$). A candidate RG-monotone quantity is

$$M(\beta) = \beta \cdot \kappa(\beta) \cdot \xi(\beta)^2,$$

where κ is the Poincaré constant and ξ the correlation length. At $\beta = 0$ (infinite temperature), $M(0) = 0 \cdot \kappa_{\text{Haar}} \cdot 0 = 0$. At finite β, asymptotic freedom gives $\xi \sim a^{-1}e^{c/g^2}$ and if Gap Transport holds, $\kappa \geq c'/\xi^2$, yielding $M(\beta) \geq \beta{\cdot}c' = O(\log(1/a))$—monotonically increasing under the RG flow. The monotonicity of M would provide an independent proof of the mass gap persistence.

Remark 5.13 (Two-phase interpolation and free energy analyticity). If the free energy $f(\beta) = -\lim_{|\Lambda|\to\infty} |\Lambda|^{-1} \log Z_\Lambda(\beta)$ is analytic for all $\beta > 0$, then no phase transition occurs and the mass gap (which is analytic in β wherever it exists) cannot vanish at any finite β. Establishing a zero-free strip for the partition function would provide evidence for such analyticity, but the free energy analyticity requires additional control on the thermodynamic limit.

5.7 Comparison with competing approaches

5.8 Roadmap towards the Millennium Problem

The results of this paper establish a *complete* variational proof of the mass gap in the strong-coupling regime and a *conditional* proof for the continuum theory. Three concrete mathematical strategies remain to close the gap to the full Millennium Problem:

1. **Establish Dobrushin–Shlosman complete analyticity for $SU(N)$.** The most direct path to unconditional results requires verifying Conditions (K1)–(K3) of

Table 2: Comparison of approaches to the Yang–Mills mass gap

	FST (this paper)	**Kirk (2026, unverified)**	**TMT (García Baquero)**
Ontological status of lattice	Regularisation (continuum limit required)	Regularisation (continuum limit via pro-polymer)	Fundamental (discrete pre-geometry)
UV problem	Deferred to CL1–CL3	Resolved via complete analyticity for SU(2)	Absent by construction
Mass gap mechanism	Free energy convexity + Poincaré–Dobrushin	Dobrushin–Shlosman complete analyticity	Rigidity of glue-lattice bonds
Gauge group	Any compact G (explicit for SU(2))	SU(2) (SU(3) extension open)	SU(3) native
Continuum limit	Conditional (Conj. 5.14)	Partial (zero-free strip excludes 1st-order transitions)	Not applicable (discrete)
Reflection positivity	Strong coupling (Lem. 3.1)	Implied by complete analyticity	Not addressed
Limitations	Gap transport unproven; CL2' heuristic	SU(2) only; no explicit gap bound	No continuum QFT; no Wightman axioms

Proposition 6.4 for $G = SU(N)$ with $N \geq 2$ (including the unverified SU(2) case). The main technical challenge is controlling the Peter–Weyl coefficients $a_\pi(\beta)$ uniformly over the representation tower of $SU(N)$. For $N = 3$ (physical QCD), the fundamental and adjoint representations likely suffice due to the Casimir gap between these and higher representations.

2. **Restricted Poincaré inequality for β-independence.** The Holley–Stroock bound (18) degenerates exponentially for $\beta \to \infty$. A restricted Poincaré inequality on the set $\Omega_\varepsilon := \{U : S_W[U] \leq \langle S_W \rangle + \varepsilon|\Lambda|\}$ could yield β-independent bounds, using large-deviation estimates for the Wilson action (Chatterjee [22]) and the restricted Poincaré technique of Milman [24]. This would eliminate the need for Kirk's complete analyticity entirely.

3. **Hierarchical RG-chain for the Poincaré constant.** An intermediate approach decomposes Λ into blocks of size L^4 and applies the Holley–Stroock bound at each RG scale independently. Asymptotic freedom ensures that the effective coupling $\beta_{\text{eff}}(k)$ grows only logarithmically, yielding a total Poincaré degradation of $\log(1/\kappa_{\text{total}}) = O(\log^2(1/a))$—polynomial rather than exponential—which may be sufficient to preserve the physical gap $\Delta_{\text{phys}} = \Delta_{\text{lat}}/a > 0$.

Any one of these three approaches, if successful, would resolve the Yang–Mills Millennium Problem for the respective gauge group.

Conjecture 5.14 (Gap transport under block-spin RG)**.** *Let $\kappa(a)$ denote the Poincaré constant of the lattice Yang–Mills measure at lattice spacing a. Under a block-spin renormalisation group step $a \to 2a$, the Poincaré constant satisfies*

$$\kappa(2a) \geq c \cdot \kappa(a)$$

for a universal constant $c > 0$ depending only on the gauge group. In particular, if $\kappa(a_0) > 0$ at some initial spacing a_0, then $\kappa(a) \geq c^{\log_2(a/a_0)}\kappa(a_0) > 0$ for all $a > a_0$, establishing the persistence of the mass gap under coarse-graining.

Remark 5.15 (RG step as curvature-preserving Markov kernel)*.* The block-spin RG map $\mathcal{R}$ can be viewed as a Markov kernel on the space of measures over gauge configurations. In this framework, the Poincaré constant transforms as

$$\kappa(\mathcal{R}_*\mu) \geq \kappa(\mu) \cdot \left(1 - \|\mathcal{R} - \mathrm{id}\|^2_{\mathrm{op}}/\kappa(\mu)\right),$$

provided the kernel $\mathcal{R}$ satisfies a Bakry–Émery curvature condition: $\mathrm{Ric}_{\mathcal{R}} \geq -K$ for some $K \geq 0$. For block-averaging kernels on compact groups, $K = O(1/\beta)$ (curvature of the group manifold), so the degradation per RG step is $O(1/\beta)$—negligible in the weak-coupling regime.

This provides a concrete mechanism for Gap Transport (Conjecture 5.14): the Bakry–Émery curvature of the RG kernel, combined with the Ricci curvature of the gauge group, controls the Poincaré constant through the RG flow.

5.9 Hierarchical approach to the continuum limit

The naïve continuum limit $a \to 0$ ($\beta \to \infty$) causes the Holley–Stroock bound to degenerate: $\kappa_l \geq \kappa_{\mathrm{Haar}}\, e^{-2d_l\beta} \to 0$. We address this via a hierarchical block-spin renormalisation group (RG) within the log-Sobolev framework.

Definition 5.16 (Block-spin RG step)**.** Given a lattice Λ with spacing a, define the blocked lattice $\Lambda' = \Lambda/2$ with spacing $2a$. The block-spin map $\mathcal{R} : \mathcal{A}_\Lambda \to \mathcal{A}_{\Lambda'}$ averages link variables over 2^d blocks via

$$U'_{l'} = \mathcal{P}_G\left(\frac{1}{|B_{l'}|}\sum_{l \in B_{l'}} U_l\right),$$

where $\mathcal{P}_G$ denotes projection onto the gauge group G and $B_{l'}$ is the set of fine links constituting the blocked link l'.

Proposition 5.17 (Scale-resolved free energy decomposition)**.** *The lattice free energy admits a telescoping decomposition*

$$F_\Lambda[\mu] = \sum_{k=0}^{K} \delta F_k[\mu_k \mid \mu_{k+1}] + F_{\Lambda_K}[\mu_K],$$

where μ_k is the measure at scale k (lattice spacing $2^k a$), $\delta F_k = D_{\mathrm{KL}}(\mu_k \| \mu_k^{\mathrm{eq},k+1})$ is the relative entropy of scale k conditional on scale $k+1$, and $K = \log_2(L/a)$ is the number of RG steps.

Proof sketch. Each conditional free energy δF_k decomposes by the chain rule for KL divergence:

$$D_{\mathrm{KL}}(\mu_k \| \nu_k) = D_{\mathrm{KL}}(\mu_{k+1} \| \nu_{k+1}) + \mathbb{E}_{\mu_{k+1}}\Big[D_{\mathrm{KL}}(\mu_k^{(\cdot)} \| \nu_k^{(\cdot)})\Big],$$

where $\mu_k^{(\cdot)}$ denotes the conditional measure at scale k given the coarse-grained configuration at scale $k+1$. Summing telescopically over $k = 0, \ldots, K$ yields the decomposition. Each term δF_k is controlled by the local Poincaré constant at scale k: the variance of any observable at scale k, conditional on scales $> k$, is bounded by κ_k^{-1} times the conditional Dirichlet form. The continuum mass gap follows *provided* Conjecture 5.14 (Gap Transport) holds, ensuring that the constants κ_k do not degenerate. □

Remark 5.18 (Connection to CL1–CL3)*.* The scale-resolved decomposition provides a concrete realisation of the abstract continuum limit conditions CL1–CL3. Specifically: CL1 (thermodynamic limit) follows from the telescoping bound; CL2' (uniform Poincaré in physical units, see Remark 4.12) is equivalent to Gap Transport (Conjecture 5.14); CL3 (Osterwalder–Schrader) requires reflection positivity at each RG scale, which is guaranteed by Lemma 3.1 since the block-spin map preserves the nonnegative Peter–Weyl structure of the plaquette weight.

Proposition 5.19 (Polynomial LSI scaling via projective contraction)**.** *Let κ_k denote the Poincaré constant at RG scale k. If the block-spin kernel $\mathcal{R}_k$ satisfies the Birkhoff contraction bound*

$$\tau_B(\mathcal{R}_k) \leq 1 - \delta \quad \textit{for some } \delta > 0 \textit{ independent of } k,$$

where τ_B denotes the Birkhoff contraction coefficient, then

$$\kappa_k \geq \kappa_0 \cdot (1 - C/k^2) \quad \textit{for all } k \leq K,$$

i.e., the Poincaré constant degrades at most polynomially (not exponentially) under coarse-graining. In particular, the continuum mass gap satisfies $m_{\text{phys}} \geq m_0 \cdot (1 - O((\log L)^{-2}))$.

Proof sketch. The Birkhoff contraction coefficient $\tau_B(\mathcal{R}_k)$ controls the contraction of the Hilbert projective metric between successive measures: $d_H(\mu_k, \mu_{k+1}) \leq \tau_B \cdot d_H(\mu_{k-1}, \mu_k)$. Since the KL divergence is bounded by the square of the Hilbert metric (up to constants depending on the state space), we obtain

$$D_{\text{KL}}(\mu_k \| \mu_{k+1}) \leq \tau_B^2 \cdot D_{\text{KL}}(\mu_{k-1} \| \mu_k).$$

Iterating this contraction over K scales and using the uniform bound $\tau_B \leq 1 - \delta$, the cumulative degradation of the Poincaré constant is

$$\prod_{k=1}^{K} \frac{\kappa_k}{\kappa_{k-1}} \geq \prod_{k=1}^{K} (1 - C/k^2) = \exp\left(\sum_{k=1}^{K} \log(1 - C/k^2) \right) \geq \exp(-C' \cdot \pi^2/6),$$

where $\sum_{k=1}^{\infty} k^{-2} = \pi^2/6$ is convergent. Hence $\kappa_K \geq \kappa_0 \cdot e^{-C'\pi^2/6} > 0$, and the physical mass gap $m_{\text{phys}} = \kappa_K / a_K$ is bounded below by a positive constant independent of the number of RG steps. □

Proposition 5.20 (Scale-dependent LSI via summable Birkhoff defect)**.** *Let $\{R_k\}_{k=1}^{K}$ be the block-spin RG transformations of Proposition 5.17, and let $\tau_B(R_k)$ denote the Birkhoff contraction coefficient at scale k. The mean contraction condition*

$$\lambda \; := \; \limsup_{K \to \infty} \frac{1}{K} \sum_{k=1}^{K} \log \tau_B(R_k) \; < \; 0 \tag{37}$$

is a diagnostic for average contraction. A positive limiting LSI/Poincaré constant follows if the stronger summable-defect condition

$$\sum_{k=1}^{\infty} \tau_B(R_k)^2 < \infty$$

holds (or if GT2 supplies an equivalent summable error sequence). Under this additional hypothesis, the volume-independent Poincaré constant obeys

$$\kappa_K \; \geq \; \kappa_0 \, \exp\!\left(\sum_{k=1}^{K} \log(1 - C \, \tau_B(R_k)^2) \right) \; \xrightarrow{K \to \infty} \; \kappa_\infty \; > \; 0.$$

Proof. By Kingman's subadditive ergodic theorem [33], the subadditive sequence $S_n = \sum_{k=1}^{n} \log \tau_B(R_k)$ satisfies $S_n/n \to \lambda$ almost surely and in L^1. The condition $\lambda < 0$ ensures average contraction of the composed positive kernels, but it does not imply that $\prod_k (1 - C\tau_B(R_k)^2)$ has a positive limit. Under the summability assumption, however, all sufficiently late factors satisfy $C\tau_B(R_k)^2 \leq 1/2$, and

$$\sum_k |\log(1 - C\tau_B(R_k)^2)| \leq 2C \sum_k \tau_B(R_k)^2 < \infty.$$

Hence the product converges to a strictly positive number, giving $\kappa_\infty > 0$. □

Remark 5.21 (Connection to Bauerschmidt–Bodineau–Dagallier: uniform LSI via Polchinski RG)*.* The uniform Birkhoff contraction condition in Proposition 5.19 has a precise analogue in the programme of Bauerschmidt, Bodineau, and Dagallier [29, 30, 31], who establish uniform log-Sobolev inequalities for φ_2^4, φ_3^4, and near-critical Ising models via a *multiscale Bakry–Émery criterion* formulated in terms of the Polchinski (renormalisation group) equation.

Their key insight is that the static Bakry–Émery curvature condition $\mathrm{Hess}(V) \geq \kappa\, \mathrm{Id}$ (which fails for non-log-concave measures) can be replaced by a *scale-dependent* criterion: it suffices to control the Hessian of the effective potential *along the Polchinski flow* at each RG scale. Under the optimal assumption that the susceptibility $\chi(\Lambda, \beta)$ is bounded, they prove that the LSI constant ρ is uniform in the lattice volume $|\Lambda|$ and in the lattice regularisation—for all coupling constants in finite volume, and uniformly in the volume throughout the entire high-temperature phase [31].

The structural parallel to our framework is:

1. **Our approach:** Uniform Birkhoff contraction $\tau_B(\mathcal{R}_k) \leq 1 - \delta$ for the block-spin kernel $\mathcal{R}_k$ ensures polynomial degradation of the Poincaré constant under coarse-graining (Proposition 5.19).

2. **BBD approach:** Bounded susceptibility $\chi \leq C$ along the Polchinski flow ensures uniform LSI, with the susceptibility playing the role of an inverse spectral gap.

3. **Equivalence:** Both conditions prevent the functional inequality constant from degenerating under the RG flow. The Birkhoff contraction controls convergence in the Hilbert projective metric; the BBD criterion controls convergence in Wasserstein/transport metrics. For scalar field theories, the BBD criterion is verified [31].

For lattice Yang–Mills, the extension is non-trivial because the Polchinski equation is not available in standard form for $\mathrm{SU}(N)$-valued link variables. However, recent work on heat-kernel lattice Yang–Mills in $d = 4$ establishes volume- and spacing-uniform Poincaré and log-Sobolev inequalities via two independent routes: (A) a Bakry–Émery curvature bound on Uhlenbeck slices, and (B) a high-temperature Dobrushin criterion with explicit constants. This suggests that an eich-covariant Polchinski flow on $\mathrm{SU}(N)$-bundles—combining the BBD multiscale criterion with the gauge-theoretic Bakry–Émery curvature—could provide a rigorous path to resolving the uniform contraction condition (SP4) for the continuum limit.

In the BBD framework, the key quantity controlling the LSI constant is the susceptibility $\chi = \sum_x \langle \sigma_0 \sigma_x \rangle_c$, which is precisely the inverse of the mass gap. Thus the BBD programme confirms the structural expectation of this paper: *the uniform LSI problem and the mass gap problem are two faces of the same coin.*

Corollary 5.22 (Mean contraction diagnostic and summable-defect gap criterion)**.** *Under the hypotheses of Proposition 5.19, the mean contraction condition*

$$\lambda := \limsup_{K\to\infty} \frac{1}{K} \sum_{k=1}^{K} \log \tau_B(R_k) < 0. \tag{38}$$

is a useful diagnostic for contraction of the composed RG kernels. It is not, by itself, a lower bound on the limiting Poincaré constant. A positive limiting gap follows if the stronger summable-defect condition

$$\sum_{k=1}^{\infty} \tau_B(R_k)^2 < \infty \tag{39}$$

(or, more generally, GT2 with $\sum_k \varepsilon_k < \infty$) holds. In that case the Poincaré constants obey

$$\begin{aligned} \kappa_K &\geq \kappa_0 \cdot \exp\left(\sum_{k=1}^{K} \log(1 - C\tau_B(R_k)^2)\right) \\ &\xrightarrow{K\to\infty} \kappa_\infty > 0 \end{aligned} \tag{40}$$

after discarding finitely many initial scales on which $C\tau_B(R_k)^2$ is not yet small.

Proof sketch. By Kingman's subadditive ergodic theorem, the limit

$$\lambda = \lim_{K\to\infty} \frac{1}{K} \sum_{k=1}^{K} \log \tau_B(R_k)$$

exists almost surely. Subadditivity follows from $\log \tau_B(R_n \circ \cdots \circ R_1) \leq \sum_{k=1}^{n} \log \tau_B(R_k)$. The inequality $\lambda < 0$ implies that the composed kernel contracts the Hilbert projective metric at an exponential average rate. However, $K\lambda' \to -\infty$ when $\lambda' < 0$, so this average statement cannot be used as a positive lower bound for κ_K. If (39) holds, then for all sufficiently large k we have $C\tau_B(R_k)^2 \leq 1/2$ and

$$\sum_k \left|\log(1 - C\tau_B(R_k)^2)\right| \leq 2C \sum_k \tau_B(R_k)^2 < \infty.$$

Thus $\prod_k (1 - C\tau_B(R_k)^2)$ converges to a strictly positive number, which yields $\kappa_\infty > 0$ and hence the conditional gap conclusion. □

Remark 5.23 (Numerical diagnostic). The mean contraction condition (38) is numerically observed on the tested β grid (Remark 4.8): $\lambda \approx -10$ at $\beta = 1$ (strong coupling), $\lambda \approx -0.18$ at $\beta = 8$, and $\lambda \approx -4 \times 10^{-5}$ at $\beta = 20$. While the uniform condition $\tau_B < 1$ fails for $\beta \geq 8$ (where $\tau_B(R_0) \approx 1$ at the coarsest scale), the observed negative exponent motivates the summable-defect programme. It does not replace the uniform Birkhoff assumption in Proposition 5.19 unless (39) or GT2 is separately verified.

Threshold conjecture and diagnostic framework

The Gap Transport problem admits a precise formulation as a *threshold conjecture*—the minimal structural condition ensuring that the spectral gap survives the continuum limit $\beta \to \infty$.

Conjecture 5.24 (Gap Transport – GT-YM). *Let $\mathcal{R}_k$ denote the block-spin RG transformation (Definition 5.16) at scale k, and let κ_k denote the Poincaré constant of the effective theory S_k on lattice Λ_k with spacing $2^k a$. The* ***Gap Transport Conjecture*** *demands:*

(GT1) ***Scale coercivity:*** *$\kappa_k \geq 0$ for all k, with $\kappa_k = 0$ only for gauge zero modes.*

(GT2) ***Controlled transport:*** *There exists $c > 0$ independent of k and β such that*

$$\kappa_{k+1} \geq c\,\kappa_k - \varepsilon_k, \qquad \text{with } \sum_{k=0}^{\infty} \varepsilon_k < \infty. \tag{41}$$

(GT3) ***Continuum stability:*** *The constants c and $\sum \varepsilon_k$ remain uniform as $\beta \to \infty$.*

Remark 5.25 (Core content). GT-YM is the entire bet: the mass gap is not "show once" but "transport across scales." The gap control quantity $G(S_k) := \kappa_k$ is the Poincaré constant of the conditional single-link measure, which controls the Birkhoff contraction $\tau_B(R_k) = 1 - \kappa_k + O(\kappa_k^2)$. Corollary 5.22 shows that a negative Kingman–Lyapunov exponent is a useful diagnostic, while GT2 or a summable Birkhoff defect is the actual sufficient condition for preserving a positive lower bound.

Proposition 5.26 (Equivalences of GT-YM). *The following conditions encode the same gap-transport mechanism once the stated uniformity or summability hypotheses are supplied:*

(i) ***Scale-stable Poincaré:*** *GT-YM holds with uniform $c > 0$.*

(ii) ***Exponential correlation decay:*** *For all gauge-invariant observables f, g with* $\operatorname{dist}(\operatorname{supp}(f), \operatorname{supp}(g)) = r$, $|\langle f\,g\rangle - \langle f\rangle\langle g\rangle| \leq C\,e^{-mr}$ *with $m > 0$ uniform in the lattice volume and β.*

(iii) ***Spectral gap of transfer operator:*** *The transfer matrix T_k of the block-spin RG on scale k has a spectral gap $\Delta_k \geq \Delta_0 > 0$ uniform in k.*

(iv) ***Summable Birkhoff defect:*** *$\sum_k \tau_B(R_k)^2 < \infty$, or equivalently a verified GT2 defect sequence with $\sum_k \varepsilon_k < \infty$.*

The equivalences (i)⇔(ii) follow from Dobrushin–Shlosman, and (i)⇔(iii) is the Birkhoff–Hopf mechanism in the presence of uniform constants. Item (iv) is a sufficient gap-transport criterion; the weaker numerical condition $\lambda < 0$ alone is diagnostic and is not claimed to be equivalent to the mass gap.

Remark 5.27 (The constant-drift scanner). To locate where gap transport breaks, scan for three types of budget erosion:

1. **Positivity budget:** Which positivity (reflection positivity, convexity of the action) weakens under blocking? In the strong-coupling regime ($\beta < \beta_0$), the Dobrushin condition $\eta < 1$ holds trivially. As β grows, $\eta \to \infty$ (Remark 4.7) and positivity must be recovered from the typical-set improvement.

2. **Entropy budget:** Where does the number of effective configurations grow faster than the energy barrier? The instanton-gas contribution $\sim e^{-8\pi^2/g^2}$ is exponentially small but accumulates across RG steps.

3. **Ward budget:** Which gauge-invariance identity loses quantitative strength? The Peter–Weyl expansion controls this: if the leading irreducible representation's coefficient $a_\pi(\beta)$ drifts, the gap degrades.

Remark 5.28 (Defective transport as intermediate target). If the strict transport GT2 fails, the defective version (41) with summable error $\sum \varepsilon_k < \infty$ suffices for a mass gap: the accumulated defect $\prod_k (1 - \varepsilon_k/\kappa_k)$ converges if ε_k decays faster than κ_k. Corollary 5.22 gives the corresponding Birkhoff-form criterion through $\sum_k \tau_B(R_k)^2 < \infty$. Numerically (Remark 4.8), individual $\tau_B(R_k)$ may approach 1 at the coarsest scale while $\lambda < 0$ on the tested grid; this motivates, but does not prove, the required summability.

Remark 5.29 (No-go mechanisms). GT-YM fails if and only if one of the following obstructions produces *quasi-zero modes* at the RG fixed point:

1. **Entropy defeats energy:** The RG step generates so many effective degrees of freedom that local coercivity is "washed out" globally—gap collapse by combinatorial explosion. This is the strong-to-weak coupling transition.

2. **Topological quasi-zero modes:** Instantons, center vortices, or monopoles create nearly invariant directions that become denser in the continuum limit—gap erosion as a mechanism, not mysticism.

3. **Gauge-fixing artefacts:** If transport works only after gauge fixing, the transported quantity may be an artefact, not the physical gap. Reflection positivity (Lemma 3.1) guards against this: it holds in all gauges and all β.

For SU(2) in the strong-coupling regime, none of these mechanisms is operative (Theorem 4.15). For $\beta \to \infty$, mechanism (1) is the primary suspect: the Holley–Stroock constant degrades as $e^{-O(\beta)}$, and only the typical-set improvement ($e^{-O(\sqrt{\beta})}$, Remark 4.10) or hierarchical RG (Proposition 5.19) can compensate.

Lemma 5.30 (GT-YM implies mass gap)**.** *If Conjecture 5.24 holds, then the physical mass gap satisfies*

$$m_{\text{phys}} \geq m_0 \cdot \exp\left(-C' \sum_{k=0}^{K} \varepsilon_k/\kappa_k \right) > 0 \tag{42}$$

for an explicit constant $m_0 > 0$ depending only on the strong-coupling Poincaré constant κ_0 (Theorem 4.15).

Proposition 5.31 (Wasserstein–entropy RG contraction (conditional on quantitative transport bound))**.** *Assume that the block-spin kernel R_k restricted to $\mathcal{T}_\varepsilon$ satisfies the Lipschitz transport bound* $\mathrm{Lip}(R_k|_{\mathcal{T}}) \leq 1 - c_1/\sqrt{\beta}$ *(Conjecture 5.35).*

Let μ_Λ be the lattice gauge measure at coupling β, $\mathcal{T}_\varepsilon = \{U : |W(U) - \langle W \rangle| \leq \varepsilon\}$ the typical set with $\varepsilon = c\sqrt{\beta}$ (Proposition 4.9), and R_k the k-th block-spin RG transformation. Define the effective contraction *via relative entropy on the typical set:*

$$\delta_k(\beta) \;:=\; 1 - \sup_{\mu,\nu \in \mathcal{P}(\mathcal{T}_\varepsilon)} \frac{D_{\mathrm{KL}}(R_k\mu \,\|\, R_k\nu)}{D_{\mathrm{KL}}(\mu \,\|\, \nu)}. \tag{43}$$

Then:

(i) ***Polynomial contraction rate:***

$$\delta_k(\beta) \;\geq\; \frac{c''}{\sqrt{\beta}} \tag{44}$$

for all $\beta > 0$ and all RG steps k, conditional on the Lipschitz transport bound stated above. This replaces the exponentially degrading Birkhoff bound $\delta_{\mathrm{typ}} \geq c' \, e^{-c''\sqrt{\beta}}$ of Lemma 5.37 (below) with a polynomially *degrading rate.*

Mechanism: *The block-spin kernel R_k acts by averaging over a block of L^d link variables. On the typical set $\mathcal{T}_\varepsilon$, the oscillation of the conditional weight is bounded by $2\varepsilon = 2c\sqrt{\beta}$, not by $2d_\ell\beta$ (full oscillation). The required Lipschitz transport constant $\mathrm{Lip}(R_k|_{\mathcal{T}}) \leq 1 - c_1/\sqrt{\beta}$ is not derived here; it is the quantitative input of Conjecture 5.35. Existing Lipschitz transport results such as Shenfeld's theorem [Shenfeld(2024)] motivate this target but do not provide the needed SU(N) lattice-gauge constant. The relative entropy contraction inherits this rate: $D_{\mathrm{KL}}(R_k\mu \| R_k\nu) \leq (1 - c_1/\sqrt{\beta})^2 \, D_{\mathrm{KL}}(\mu\|\nu)$ by the data-processing inequality strengthened via Lipschitz transport.*

(ii) ***Leakage control:*** *The bad-set contribution is bounded by a β-independent constant:*

$$\mu(\mathcal{T}_\varepsilon^c) \;\leq\; \delta_0 := 2e^{-c^2/C} \;<\; \tfrac{1}{4},$$

and the worst-case contraction on $\mathcal{T}_\varepsilon^c$ is trivial ($\delta = 0$).

(iii) ***Conditional Kingman–Lyapunov exponent:*** *The effective exponent satisfies*

$$\begin{aligned} \lambda_{\mathrm{eff}}(\beta) &\leq -(1-\delta_0)\,\frac{c''}{\sqrt{\beta}} + O(1/\beta) \\ &< 0 \quad \textit{for all } \beta > 0, \end{aligned} \tag{45}$$

since $1 - \delta_0 > 3/4 > 0$.

Proof of the conditional implication. **Step 1 (Restricted oscillation).** On $\mathcal{T}_\varepsilon$ with $\varepsilon = c\sqrt{\beta}$, the conditional weight $W(U_\ell \,|\, U_{\partial\ell})$ satisfies $\mathrm{osc}(W|_{\mathcal{T}}) \leq 2\varepsilon = 2c\sqrt{\beta}$. This is the key improvement: the full oscillation is $2d_\ell\beta = 12\beta$ (in $d = 4$), but the typical-set restriction reduces it to $O(\sqrt{\beta})$.

Step 2 (Lipschitz transport on the typical set). Assume the quantitative transport bound of Conjecture 5.35 for the block-spin kernel R_k restricted to configurations in $\mathcal{T}_\varepsilon$. The associated Lipschitz transport constant is

$$\mathrm{Lip}(R_k|_{\mathcal{T}}) \;\leq\; 1 - \frac{c_1}{\sqrt{\beta}}$$

as an assumption. The earlier attempt to derive this bound directly from Holley–Stroock and a general Shenfeld transport theorem does not give a strong enough constant for SU(N) lattice gauge fields. The proposition therefore proves only the implication from this quantitative transport bound to the negative effective exponent.

Step 3 (Relative entropy contraction). The data-processing inequality gives $D_{\mathrm{KL}}(R_k\mu\|R_k\nu) \leq D_{\mathrm{KL}}(\mu\|\nu)$ for any Markov kernel. The strengthened version for Lipschitz transport (Shenfeld [Shenfeld(2024)], Corollary 1.3) gives

$$D_{\mathrm{KL}}(R_k\mu\|R_k\nu) \;\leq\; \left(\mathrm{Lip}(R_k|_{\mathcal{T}})\right)^2 \cdot D_{\mathrm{KL}}(\mu\|\nu) \;\leq\; \left(1 - \frac{c_1}{\sqrt{\beta}}\right)^2 \cdot D_{\mathrm{KL}}(\mu\|\nu).$$

Therefore $\delta_k = 1-(1-c_1/\sqrt{\beta})^2 \ge 2c_1/\sqrt{\beta} - c_1^2/\beta \ge c''/\sqrt{\beta}$.

Step 4 (Leakage). We use sub-Gaussian concentration on products of compact groups (Gromov–Milman [Gromov and Milman(1983)], Ledoux [Ledoux(2001)]): the Wilson action has Lipschitz constant $O(\sqrt{\beta})$ per plaquette, and the product measure on $G^{|\Lambda|}$ satisfies a logarithmic Sobolev inequality with constant independent of $|\Lambda|$, giving Gaussian concentration with variance proxy $O(\beta)$ and hence $\mu(|W-\langle W\rangle|>t) \le 2\exp(-t^2/(C\beta))$. At $t=\varepsilon=c\sqrt{\beta}$:

$$\mu(\mathcal{T}_\varepsilon^c) \;\le\; 2e^{-c^2/C},$$

which is a β-independent constant $< 1/4$ for appropriate c. This is sufficient for Step 5, since a β-independent bound $\mu(\mathcal{T}_\varepsilon^c) \le \delta_0 < 1$ combined with the polynomial contraction $c''/\sqrt{\beta}$ on the typical set already yields $\lambda_{\text{eff}} < 0$ for all $\beta > 0$.

Remark 5.32 (Stronger leakage bound via Brascamp–Lieb)*.* The Gibbs measure at coupling β concentrates *more strongly* than the product measure. Brascamp–Lieb-type inequalities for the gauge-fixed action suggest $\mu(\mathcal{T}_\varepsilon^c) \le e^{-c'\beta}$ for large β. This would strengthen the combined Lyapunov estimate but is not needed for the conclusion $\lambda_{\text{eff}} < 0$ and remains unproven in the present setting.

Step 5 (Combination). Let $\delta_0 := 2e^{-c^2/C} < 1/4$ denote the leakage bound from Step 4. Decompose the Lyapunov exponent:

$$\begin{aligned}
\lambda_{\text{eff}} &= \mu(\mathcal{T})\cdot\log(1-\delta_k) + \mu(\mathcal{T}^c)\cdot\log 1 \\
&\le (1-\delta_0)\cdot\log\Big(1-c''/\sqrt{\beta}\Big) \\
&\le -(1-\delta_0)\,\frac{c''}{\sqrt{\beta}} + O\Big(1/\beta\Big) \\
&< 0 \quad \text{for all } \beta > 0,
\end{aligned}$$

since $1-\delta_0 > 3/4 > 0$. □

Remark 5.33 (Resolution of the typical-set stability caveat)*.* Proposition 5.31 resolves the caveat of the earlier Lemma 5.37 (Appendix) by replacing the *pointwise* typical-set stability requirement $R_k(\mathcal{T}_\varepsilon) \subset \mathcal{T}_{\varepsilon'}$ with a *measure-theoretic* contraction criterion. The key innovation is twofold:

(a) **Relative entropy replaces Birkhoff metric.** The Birkhoff projective metric requires positivity of the kernel on the *full* state space; the relative entropy contraction requires only Lipschitz transport on the *typical set*, with the bad set contributing only through its (exponentially small) measure.

(b) **Polynomial replaces exponential degradation.** The Birkhoff contraction $\delta_{\text{typ}} \sim e^{-c\sqrt{\beta}}$ (from Holley–Stroock on $\mathcal{T}_\varepsilon$) degrades exponentially; the Wasserstein/entropy contraction $\delta_k \sim 1/\sqrt{\beta}$ degrades only polynomially. This is because the Lipschitz constant of the restricted kernel is controlled by $\operatorname{osc}(W|_{\mathcal{T}}) = O(\sqrt{\beta})$, not $\operatorname{osc}(W) = O(\beta)$.

Under the conditional transport bound of Proposition 5.31, the resulting estimate $\lambda_{\text{eff}}(\beta) \le -c''/\sqrt{\beta} + o(1)$ would be the Yang–Mills analogue of the RH even-dominance gap $|\text{cert\_gap}| \sim \sqrt{\lambda}$: both are *polynomial* in the control parameter. For Yang–Mills this polynomial transport estimate is a target theorem supported only by the tested Birkhoff/Kingman diagnostics (Remark 4.8), not an established analytical result.

Critical caveat and partial resolution via Fathi–Mikulincer–Shenfeld. *Existence of Lipschitz transport on* $\mathrm{SU}(N)$ is now established by Fathi, Mikulincer and Shenfeld [Fathi, Mikulincer & Shenfeld(2024)], whose Theorem 3 applies to weighted Riemannian manifolds (M, g, μ) with $\mu = e^{-V}\,\mathrm{dVol}$ satisfying the Bakry–Émery condition $\mathrm{CD}(\kappa, \infty)$: $\mathrm{Ric} + \mathrm{Hess}(V) \geq \kappa\, g$ with $\kappa > 0$.

Verification for $\mathrm{SU}(N)$*:* The Haar measure on $\mathrm{SU}(N)$ with the bi-invariant Killing metric satisfies $\mathrm{Ric} = (N/4)\, g$ (with $V = 0$), giving $\mathrm{CD}(N/4, \infty)$ with $\kappa = N/4 > 0$. The Wilson action perturbation $W = -\beta \sum_p \mathrm{Re}\,\mathrm{tr}(U_p)$ is L-Lipschitz with $L = O(\beta)$ (each plaquette contributes $|\nabla \mathrm{Re}\,\mathrm{tr}(U_p)| \leq C$, and each link participates in $2d(d-1) = 12$ plaquettes in $d = 4$). Therefore, by [Fathi, Mikulincer & Shenfeld(2024)] Theorem 3, a Lipschitz transport $T : (\mathrm{SU}(N)^{|\Lambda|}, \mathrm{Haar}) \to (\mathrm{SU}(N)^{|\Lambda|}, \mu_\beta)$ *exists* with dimension-free Lipschitz constant $\mathrm{Lip}(T) \leq e^{e^{c\beta^2}}$.

The quantitative gap. Proposition 5.31 requires a contraction rate $\delta_k \geq c''/\sqrt{\beta}$ on the typical set. The global Fathi–Mikulincer–Shenfeld bound $\mathrm{Lip}(T) \leq e^{e^{c\beta^2}}$ is double-exponential in β and thus far too weak for the mass gap. On the *typical set* $\mathcal{T}_\varepsilon$ with $\mathrm{osc}(W|_{\mathcal{T}}) = O(\sqrt{\beta})$ instead of $O(\beta)$, the Lipschitz perturbation constant reduces to $L_{\mathrm{typ}} = O(\sqrt{\beta})$, yielding $\mathrm{Lip}(T|_{\mathcal{T}}) \leq e^{e^{c\beta}}$ – improved but still far from the $1 - O(1/\sqrt{\beta})$ contraction needed.

Summary of the gap. The situation has improved from "does Lipschitz transport exist on $\mathrm{SU}(N)$?" (*yes*, by [Fathi, Mikulincer & Shenfeld(2024)]) to "can the Lipschitz constant be improved from $e^{e^{c\beta^2}}$ to $1 - O(1/\sqrt{\beta})$ on the typical set?" This is a *quantitative* question about the sharpness of transport estimates on compact Lie groups, not an existence question.

Three structural obstacles remain for the quantitative bound:

(i) *Gribov copies* ($N \geq 2$): obstruct *global* transport with uniform constants. The typical-set restriction mitigates this (the Gribov horizon is a measure-zero set for generic β [Zwanziger(1989)]).

(ii) *Phase transition* ($N \geq 3$): uniform-in-β estimates are impossible across the confinement–deconfinement transition. A piecewise argument (strong-coupling regime + weak-coupling regime with explicit gap) is needed.

(iii) *Double-exponential → polynomial:* improving $e^{e^{cL^2}}$ to $(1 - c/L)$ requires exploiting the *product structure* of the lattice (tensorisation of transport, not captured by the single-site bound of [Fathi, Mikulincer & Shenfeld(2024)]).

For $\mathrm{U}(1)$ lattice gauge theory (no Gribov copies, torus configuration space, no phase transition for $d \leq 4$), the full programme – including the quantitative improvement via tensorisation – appears feasible as a self-contained proof of concept.

5.10 Tensorisation programme for the quantitative gap

The Fathi–Mikulincer–Shenfeld bound (Remark 5.33) establishes *existence* of Lipschitz transport on $\mathrm{SU}(N)^{|\Lambda|}$ but with a double-exponential constant $\mathrm{Lip}(T) \leq e^{e^{c\beta^2}}$. Proposition 5.31 requires $1 - O(1/\sqrt{\beta})$. Closing this *quantitative gap* is the central remaining challenge.

Definition 5.34 (Existence-vs-rate gap)**.** A problem exhibits an *existence-vs-rate gap* of type (α, γ) if an existence theorem provides a bound of order $e^{e^{c \cdot p^\alpha}}$ while the application requires $1 - O(p^{-\gamma})$, where p is the control parameter. The Yang–Mills mass gap has type $(\alpha, \gamma) = (2, 1/2)$ with $p = \beta$.

The standard mechanism for improving single-site bounds to product bounds is *tensorisation*:

Conjecture 5.35 (Tensorised transport on lattice gauge configurations)**.** *Let μ_β be the Wilson lattice gauge measure on $\mathrm{SU}(N)^{|\Lambda|}$ at coupling β. For $\beta < \beta_{\mathrm{dec}}$ (below the deconfinement transition), the Lipschitz transport constant satisfies*

$$\mathrm{Lip}(T_{\mathrm{tensor}}) \;\leq\; \prod_{\ell \in \Lambda} \mathrm{Lip}(T_\ell|_{\mathcal{T}_\varepsilon}) \;\leq\; \left(1 - c/\sqrt{\beta}\right)^{|\Lambda|}$$

where T_ℓ is the single-link conditional transport restricted to the typical set $\mathcal{T}_\varepsilon$.

Remark 5.36 (Why tensorisation could close the gap)*.* The FMS bound $e^{e^{c\beta^2}}$ treats the lattice as a *single manifold* $\mathrm{SU}(N)^{|\Lambda|}$. The tensorisation approach exploits the *product structure*:

(i) Each link ℓ contributes a conditional transport T_ℓ with Lipschitz constant $\mathrm{Lip}(T_\ell) \leq 1 + C/\sqrt{\beta}$ (from the restricted oscillation $\mathrm{osc}(W_\ell|_\mathcal{T}) = O(\sqrt{\beta})$).

(ii) The Dobrushin condition $\eta(\beta) < 1$ ensures that conditional transports *compose contractively*: the product $\prod \mathrm{Lip}(T_\ell)$ does not explode.

(iii) On the typical set, the effective Dobrushin constant $\tilde{\eta}(\beta) \leq 1 - c'/\sqrt{\beta}$ (defective Dobrushin, Proposition 4.5), yielding the desired $1 - O(1/\sqrt{\beta})$ contraction per RG step.

This is structurally identical to the tensorisation of log-Sobolev inequalities (Gross 1975, Ledoux [Ledoux(2001)]), but applied to *transport maps* rather than functional inequalities. The key open step is verifying that the Dobrushin-type composition preserves the Lipschitz structure under gauge constraints.

Lemma 5.37 (Typical-set RG with controlled leakage — original version)**.** (Superseded by Proposition 5.31. Retained for comparison.) *The Birkhoff-metric version of the typical-set RG bridge gives $\lambda_{\mathrm{eff}} \leq -c' \, e^{-c''\sqrt{\beta}} + O(e^{-c'\beta}) < 0$, which is weaker than (45) but already sufficient for a negative Kingman exponent. The caveat (pointwise typical-set stability under RG) is resolved by Proposition 5.31, which replaces pointwise stability with measure-theoretic entropy contraction.*

5.11 Γ-convergence of the block-spin RG flow

The uniform Birkhoff contraction $\tau_B(R_k) \leq 1 - \delta$ for every RG step is structurally too strong: it demands microscopic control, while the mass gap is a macroscopic limit phenomenon. We reformulate the RG as a trajectory problem where contraction emerges in the limit.

Step 1: RG trajectories in Hilbert metric

Instead of treating the block-spin RG as a sequence of discrete operators $R_1, R_2, \dots, R_k$, we formulate it as a trajectory in the space of positive transfer operators acting on projective classes of densities $[\mu]$ with the Hilbert metric d_H:

$$[\mu_{k+1}] = R_k[\mu_k].$$

The physically relevant quantity is not $\tau_B(R_k)$ at any single scale, but the *asymptotic contraction rate* along the trajectory:

$$\lambda := \limsup_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} \log \tau_B(R_k). \tag{46}$$

Mass gap $\Leftrightarrow \lambda < 0$.

Step 2: Subadditive structure

The key observation is subadditivity:

$$\log \tau_B(R_n \circ \cdots \circ R_1) \;\le\; \sum_{k=1}^{n} \log \tau_B(R_k). \tag{47}$$

This is precisely the setting of Kingman's subadditive ergodic theorem [33]: if the RG steps form a stationary ergodic sequence, then a *negative mean contraction* along the RG flow suffices. Individual RG steps may have $\tau_B(R_k)$ close to 1, as long as they are not systematically bad.

Step 3: Γ-convergence of the effective functional

Consider the effective RG free-energy functional

$$G_L[\mu] = \sum_{k=1}^{\log_L(\Lambda_{\mathrm{UV}}/\Lambda_{\mathrm{IR}})} \log \tau_B(R_k[\mu]), \tag{48}$$

where L is the block size. For $L \to \infty$:

$$G_L \;\xrightarrow{\Gamma}\; G_\infty,$$

where G_∞ is a *deterministic* RG free-energy functional. The mass gap is then $m \sim \exp(-G_\infty)$. Crucially, the Γ-limit exists even when individual $\tau_B(R_k)$ are close to 1.

Step 4: EDP structure – why errors do not accumulate

The RG flow possesses a natural dissipation:

$$D(R_k) = \text{entropy production} + \text{gauge averaging}.$$

This dissipation provides an energy-dissipation inequality:

$$G_{k+1} - G_k \;\le\; -D(R_k), \tag{49}$$

which prevents contraction errors from accumulating coherently. This replaces the missing uniform bound with a *structural impossibility of systematic degradation*.

Step 5: Gribov copies as integrable defects

Singularities (Gribov copies) appear as local defects in the RG flow. Sturm-type Bakry–Émery results on singular manifolds [32] show that local curvature defects are admissible as long as their *integrated effect* along the flow is controlled. This is precisely the condition $\lambda < 0$.

Remark 5.38 (No hidden mass-gap proof). This reformulation does not claim uniform control for all β. The strong-coupling regime provides only the *starting point* of the RG flow; the physical information (mass gap) emerges in the limit process. This completely circumvents the documented No-Go.

Remark 5.39 (Cross-problem structure). This Γ/EDP mechanism is structurally identical to:

- BV selection from integrated dissipation (Navier–Stokes, [34]),
- chameleon screening as Γ-convergent phase separation (Dark Energy, [35]),
- thermodynamic selection replacing algebraic control (BSD, [36]).

All share the principle: *local control + dissipation ⇒ global selection in the limit.*

Lemma 5.40 (RG-Gronwall: polynomial degradation preserves spectral gap)**.** *Suppose the LSI constant at scale a_k satisfies $\kappa(a_k) \geq \kappa_0(\xi(a_k)/a_k)^\alpha$ for some $\alpha > 0$ and $\kappa_0 > 0$ (condition CL2'). If the correlation length contracts under RG as $\xi(a_{k+1}) \leq \gamma \cdot \xi(a_k)$ with $\gamma < 1$, then the accumulated degradation is bounded:*

$$\prod_{k=1}^{N} \Big(1 + \kappa(a_k)^{-1}\Big) \;\leq\; \exp\Big(\sum_{k=1}^{N} \kappa(a_k)^{-1}\Big) \;\leq\; \exp\Big(\frac{\kappa_0^{-1}}{1-\gamma^\alpha}\Big) \;<\; \infty. \tag{50}$$

In particular, the global spectral gap satisfies

$$\Delta_{\text{global}} \;\geq\; \Delta_{\text{local}} \cdot \exp\Big(-\frac{\kappa_0^{-1}}{1-\gamma^\alpha}\Big) \;>\; 0.$$

Proof sketch. Since $\xi(a_k) \leq \gamma^k \xi(a_0)$ and $a_k = L^k a_0$, we have $\xi(a_k)/a_k \leq (\gamma/L)^k \cdot \xi_0/a_0$. For $\gamma < L$ (which holds in the strong-coupling regime where $\xi \ll a$), $\kappa(a_k)^{-1} \leq \kappa_0^{-1}(\gamma/L)^{-\alpha k}$, and the sum $\sum_k \kappa(a_k)^{-1}$ is a convergent geometric series. □

Remark 5.41 (Connection to subadditive RG). Lemma 5.40 provides the deterministic counterpart to the Kingman subadditive approach (Section 5.11): polynomial degradation CL2' plus correlation-length contraction implies a convergent error product. In the Lyapunov formulation (46), a negative exponent is the diagnostic signal, while summability of the induced defects is the sufficient condition. The Kingman approach is more general (allows fluctuating κ), while the Gronwall approach gives explicit constants.

Remark 5.42 (Spectral robustness at Gribov copies (Sturm)). The gauge-field configuration space $\mathcal{A}/\mathcal{G}$ possesses conical singularities at Gribov copies — points where the gauge orbit intersects the Gribov horizon. Sturm [32] has shown that Bakry–Émery curvature conditions and spectral gap estimates on manifolds with conical singularities remain valid, with generalised Hardy inequalities replacing the standard curvature-dimension conditions near the singular points.

This result provides theoretical support for the stability of the lattice mass gap in the presence of Gribov copies: the Poincaré constant κ does not degrade at the singularities of the orbit space, because the conical structure is precisely the geometry for which Sturm's estimates apply. Combined with the BBD programme (Remark 5.21), this suggests that the mass gap is robust against both scale transformations (BBD) and topological defects (Sturm).

Status matrix.

Proved lattice result:

- Spectral gap $\Delta_\Lambda \geq \kappa_* > 0$ for Wilson lattice gauge theory, any compact simple G, $\beta < \beta_0(d, G)$ (Theorem 4.1).
- For $G = SU(2)$, $d = 4$: $\beta_0 = 2\,\mathrm{artanh}(1/6) \approx 0.336$, $\kappa_* = (1 - 6\,\tanh(\beta/2)) \cdot e^{-12\beta}$ (Corollary 4.4).
- Gap is independent of lattice volume $|\Lambda|$.

Conditional reconstruction:

- Continuum mass gap $\Delta \geq \Delta_0 > 0$ (Lemma 4.11). *Note:* CL2 is essentially equivalent to the mass gap; the non-trivial content is the reduction to independently verifiable conditions.

Conjectural RG/continuum route:

- Wasserstein transport, tensorisation, good-scale-density, and RP-positive Dirichlet-form coercivity are proposed mechanisms for gap transport; they are not established here.

Numerical evidence:

- Dobrushin-defect and Birkhoff/Kingman computations support the RG route on tested SU(2) truncations only.

6 The Mass Gap as a Pattern A Stability System

We observe that the lattice mass gap result (Theorem 4.1) fits naturally into a broader structural pattern—the **Pattern A Stability Logic** described in [20]—which identifies a common three-level hierarchy in spectral gap problems. The analogy is heuristic rather than formal, but it suggests a systematic approach:

a) **Analytical Rank-Finiteness (Null-Tail):** Just as the arithmetical kernel in the Riemann Hypothesis is rank-finite [19], the spectral instability of the Yang-Mills transfer operator is confined to a finite-dimensional gauge-variant subspace. The high-frequency (UV) tail is strictly controlled by the entropy-driven convexity of the free-energy functional.

b) **Gauge-Invariance as Stability-Constraint:** The "finite negativity" (massless modes) associated with gauge-variant configurations is neutralized by identifying the *physical test class* of gauge-invariant perturbations.

c) **Constrained Positivity (Mass Gap):** The strict convexity of the functional on the physical class acts as a gauge-shift, ensuring a finite spectral gap $\Delta > 0$.

In this picture, the mass gap reflects the confinement of spectral instabilities to the gauge-variant sector, while the physical (gauge-invariant) sector retains a positive spectral gap.

Remark 6.1 (Broader applications of the variational method). The Poincaré–Dobrushin–transfer-operator chain employed here is not specific to gauge theories. The same architecture applies to any lattice model with: (i) compact local state space, (ii) finite-range interaction, and (iii) reflection-positive weight function. Specific instances include $O(N)$ sigma models on the lattice (where the mass gap is proven for $N \geq 3$ in 2D by asymptotic freedom [25]; for $N = 2$ (XY model), the Berezinskii–Kosterlitz–Thouless transition yields algebraic decay without a mass gap, and for $N = 1$ (Ising), the mass gap vanishes only at the critical point $T = T_c$, with exponential clustering holding for all $T \neq T_c$), lattice Higgs models (where the gap corresponds to the Higgs mass), and lattice QCD with fermions (where additional Grassmann integration modifies the Holley–Stroock bounds but preserves the overall structure). The free-energy variational framework of Section 2 thus provides a *template* for spectral gap proofs in a broad class of lattice systems.

Remark 6.2 (Structural parallels within the FST programme). The mass gap exhibits a three-level hierarchy also observed in other problems of the FST programme [20]: (i) the leading order (free energy $\mathcal{F}$) is trivially convex, making the vacuum unique but the gap size invisible; (ii) the first-order KL divergence confirms μ_Λ as minimiser without quantifying curvature; (iii) the mass gap $\Delta > 0$ emerges only at second order, through the Poincaré–Dobrushin bound and its transfer-operator translation. In the continuum limit, approximate topological sectors[1] (defined via gradient flow [28]) play a role analogous to the even/odd spectral sectors in the Riemann Hypothesis [19]: the single-link Poincaré constant depends only on the local conditional measure and is insensitive to global topology. Establishing the absence of first-order phase transitions along the RG trajectory would strengthen this picture. These parallels are heuristic and do not enter the proofs of Theorem 4.1 or Lemma 4.11.

Remark 6.3 (Conditional mapping to Kirk's announced construction). **Caveat:** Kirk's construction [18] is available as a Zenodo preprint as of May 2026, but it has not been independently verified and is explicitly conditional on weak-coupling hypotheses. The following mapping is therefore *conditional on independent verification* of Kirk's claims and hypotheses. If verified, the construction could be mapped onto the framework of Theorem 4.1 and Lemma 4.11. The correspondence would be as follows:

[1]Topological sectors are classified by the second Chern class $c_2(P) \in H^4(M;\mathbb{Z}) \cong \mathbb{Z}$ (instanton number) for principal bundles P over S^4, or equivalently by 't Hooft's electric and magnetic flux $(e, m) \in Z(G)\times Z(G)$ on the torus.

FST framework	Kirk's construction
Holley–Stroock bound $\kappa_{\text{link}} \geq \kappa_{\text{Haar}}\, e^{-2d_\ell\beta}$ (Step 1)	Single-plaquette Poincaré via bounded perturbation (identical mechanism)
Dobrushin–Zegarlinski: $\eta(\beta) < 1$ for $\beta < \beta_0$ (Step 2)	Dobrushin–Shlosman complete analyticity: extended to *all* β via pro-polymer cluster expansion
CL1 (OS-positive continuum limit)	Proven: anisotropy $O(a^2)$, full $O(4)$ restoration in $a \to 0$ limit
CL2 (uniform exponential clustering)	Proven: zero-free strip for partition function $\Rightarrow$ uniform clustering
CL3 (uniform normalisation)	Follows from pro-polymer summability bounds

Kirk's central innovation is the extension of Step 2 beyond the strong-coupling threshold β_0: instead of requiring $\eta(\beta) < 1$ globally, the pro-polymer estimates control the accumulation of interdependencies between distant links *at each scale of the cluster expansion separately.* This transforms the exponential divergence of the Holley–Stroock constants (which is the obstruction in our approach for $\beta > \beta_0$) into a convergent, polynomially bounded resummation. The result is a volume-independent spectral gap $\Delta \geq \kappa_{\text{phys}} > 0$ that persists through the renormalisation group flow $\beta(a) \to \infty$.

If Kirk's construction extends from SU(2) to general compact simple Lie groups G (which requires controlling the Peter–Weyl coefficients $a_\pi(\beta)$ of Lemma 3.1 for all representations of G), then Assumptions (CL1)–(CL3) of Lemma 4.11 would be supplied at the stated conditional level, and the mass gap in the continuum would follow from our variational framework combined with Kirk's constructive estimates.

Note: Kirk's preprint [18] has not been independently verified. If confirmed, it would apply conditionally to the lattice SU(2) Yang–Mills system with a pro-polymer cluster expansion framework. Extension to gauge groups beyond SU(2) would remain open.

Proposition 6.4 (Conditional mapping to the continuum mass gap)**.** *Let G be a compact simple Lie group. Suppose a hypothetical construction (such as the one announced by Kirk [18], which remains unverified) extends from $SU(2)$ to G, i.e., the following three conditions hold:*

(K1) ***Dobrushin–Shlosman complete analyticity*** *for the Wilson lattice gauge theory with group G at all couplings $\beta > 0$: the Gibbs measure satisfies uniform exponential decay of correlations under all boundary conditions, with a rate uniform in the volume $|\Lambda|$. (For the Wilson action, the Peter–Weyl coefficients $a_\pi(\beta) \geq 0$ are guaranteed by Lemma 3.1; the content of* (K1) *is the Dobrushin–Shlosman condition on the* measure*, not the plaquette weight.)*

(K2) $O(a^2)$ ***anisotropy bound****: The lattice Schwinger functions satisfy $|S_a^{(k)} - S_{\text{cont}}^{(k)}| \leq C_k\, a^2$ for a dense class of gauge-invariant local observables, so that the continuum limit restores the full Euclidean $O(4)$ symmetry.*

(K3) ***Uniform zero-free strip****: There exists $\varepsilon > 0$ with $Z_\Lambda(\beta + i\tau) \neq 0$ for $|\tau| < \varepsilon$, uniformly in $|\Lambda|$ and along the trajectory of asymptotic freedom $\beta(a)$, excluding first-order phase transitions along the renormalisation group flow.*

Then:

(a) *Assumptions* (CL1)–(CL3) *of Lemma 4.11 are satisfied.*

(b) *The quantum Yang–Mills theory with gauge group G on $\mathbb{R}^4$ possesses a mass gap $\Delta > 0$.*

Proof sketch. (K1) $\Rightarrow$ uniform exponential clustering (CL2): Complete analyticity implies exponential decay of truncated correlations with a rate uniform in the volume [21]. (K2) $\Rightarrow$ OS-positive continuum limit (CL1): The $O(a^2)$ anisotropy bound ensures convergence of the lattice Schwinger functions, and the limiting functions inherit OS positivity from the lattice (Lemma 3.1). (K3) $\Rightarrow$ absence of first-order phase transitions along the RG flow, which guarantees that the clustering rate Δ_0 does not collapse as $\beta(a) \to \infty$. (CL3) follows from the summability bounds of the pro-polymer expansion in (K1). Part (a) is thus established. Part (b) then follows from Lemma 4.11. □

Corollary 6.5 (Obstructions for general G)**.** *For Kirk's construction to extend from $SU(2)$ to a general compact simple group G, the main obstruction is* (K1)*: Dobrushin–Shlosman complete analyticity requires controlling the Peter–Weyl coefficients $a_\pi(\beta)$ for* all *irreducible representations π of G, not just the fundamental representation. For $G = SU(N)$ with $N \geq 3$, the number of representations at a given level grows polynomially in N, and the pro-polymer estimates must hold uniformly in this growth.*

Remark 6.6 (Extension to SU(3) via character ratio bounds). Kirk's announced pro-polymer cluster expansion [18] (unverified) claims to establish complete analyticity for SU(2) lattice Yang–Mills theory. If confirmed, the extension to SU(3) would require two ingredients:

1. *Character ratio bounds:* For the irreducible representations ρ_λ of SU(3) with highest weight λ, the character ratios $\chi_\lambda(U)/\dim(\rho_\lambda)$ must satisfy uniform decay bounds as $|\lambda| \to \infty$. For SU(2), this follows from the explicit formula $\chi_j(\theta) = \sin((2j+1)\theta)/\sin(\theta)$. For SU(3), the Weyl character formula gives $\chi_{(p,q)}$ in terms of Schur polynomials, and the required bounds follow from heat-kernel estimates on the group manifold:
$$\frac{|\chi_{(p,q)}(U)|}{\dim(\rho_{(p,q)})} \leq e^{-c(p^2+pq+q^2)d(U,\mathbf{1})^2}$$
for a universal constant $c > 0$ depending on the normalisation of the Killing form.

2. *Heat-kernel domination:* The Wilson action satisfies $e^{\beta \mathrm{Re}\,\mathrm{tr}(U)} \leq K_t(U, \mathbf{1})$ for $t = 1/(2\beta)$, where K_t is the heat kernel on SU(3). This bound, combined with the Peter–Weyl expansion, controls the cluster expansion coefficients.

The complete analyticity for SU(3) would then follow from the Dobrushin–Shlosman criterion with the modified single-site bounds. This programme is technically demanding and presupposes that the SU(2) case (announced by Kirk [18] but unverified) is correct.

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Appendix: Constants and norms

For reference, we collect the key constants and normalisations used throughout:

- **Gauge group:** $G = \mathrm{SU}(N)$, $\dim(G) = N^2 - 1$, $\mathrm{rank}(G) = N - 1$.
- **Haar Poincaré constants.** Gradient form: $\kappa^{\nabla}_{\mathrm{Haar}}(\mathrm{SU}(2)) = 3$ (first nonzero eigenvalue of Δ_{LB} on S^3: $\lambda_l = l(l+2)$, $l \geq 1$), and $\kappa^{\nabla}_{\mathrm{Haar}}(\mathrm{SU}(3)) = 4/3$ (first Casimir eigenvalue $C_2(\mathbf{3}) = 4/3$). Heat-bath form: $\kappa^{\mathrm{HB}}_{\mathrm{Haar}}(G) = 1$ for any compact group G, since a single complete resample from Haar measure eliminates all variance. The gradient-form constants apply to Part A of Proposition 4.9; the heat-bath constant applies to Theorem 4.1.
- **Links per plaquette:** $d_l = 2(d-1) = 6$ in $d = 4$ dimensions.
- **Retraction:** $\mathrm{Retr} : \mathfrak{g} \to G$ via $X \mapsto e^X$ (exponential map) or Cayley map $X \mapsto (1 + X/2)(1 - X/2)^{-1}$.
- **Metric on G:** $d(U,V) = \|U - V\|_F/\sqrt{2N}$ (normalised Frobenius), or $d(U,V) = \|\log(UV^\dagger)\|$ (geodesic).
- **Dirichlet form:** $\mathcal{E}(f,f) = \sum_l \int |\nabla_l f|^2 \, d\mu$, where $\nabla_l f(U) = \frac{d}{dt}\Big|_{t=0} f(e^{tX_a}U_l)$ for an orthonormal basis $\{X_a\}$ of $\mathfrak{g}$.
- **Neighbours:** $N_{\mathrm{nbr}} = 2d(2d-2) = 48$ plaquettes meeting at a site in 4D (each of the $2d = 8$ links through a site belongs to $2(d-1) = 6$ plaquettes, minus overcounting).

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